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# UV irradiation alters TFAM binding specificity and compaction of DNA

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## eLife Assessment

Mitochondrial DNA (mtDNA) exhibits a degree of resistance to mutagenesis under genotoxic stress, and this study on the mitochondrial Transcription Factor A (TFAM) presents **important** data concerning the possible mechanisms involved. The presented data are **solid**, technically rigorous, and consistent with established literature findings. The experiments are well-executed, providing **convincing** evidence on the change of TFAM-DNA interactions following UVC irradiation.

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## Abstract

Mitochondria lack nucleotide excision repair; however, mitochondrial DNA (mtDNA) is resistant to mutation accumulation following DNA damage. These observations suggest additional damage sensing or protection mechanisms. Transcription Factor A, Mitochondrial (TFAM) compacts mtDNA into nucleoids and binds differentially to certain forms of DNA damage. As such, TFAM has emerged as a candidate for protecting mtDNA or sensing damage. To examine the possibilities that TFAM might protect DNA from damage or act as a damage sensing protein for irreparable forms of mtDNA damage, we used live-cell imaging and HeLa cell-based assays, atomic force microscopy (AFM), and high-throughput protein-DNA binding assays to characterize the binding properties of human TFAM to ultraviolet-C (UVC) irradiated DNA and the cellular consequences of UVC irradiation. Our cell data show increased TFAM mRNA after exposure and suggest an increase in mtDNA degradation without a loss in mitochondrial membrane potential that might trigger mitophagy. Our protein-DNA binding assays indicate a reduction in sequence specificity of TFAM following UVC irradiation and a redistribution of TFAM binding throughout the mitochondrial genome. Our AFM data show increased compaction of DNA by TFAM in the presence of damage. Despite the TFAM-mediated compaction of mtDNA *in vitro*, we do not observe any protective effect of increased TFAM protein on DNA damage formation in cells or *in vitro*. Increased TFAM protein did not alter levels of mtDNA damage over time after UVC exposure *in vivo*, but knockdown of TFAM did alter mtDNA damage levels in HeLa cells both at baseline and after UVC exposure. Taken together, these studies indicate that UVC-induced DNA damage alters TFAM binding and promotes

compaction by TFAM *in vitro*. We hypothesize that that TFAM may act as a damage sensing protein *in vivo*, sequestering damaged genomes to prevent mutagenesis by facilitating removal or suppression of replication.

## Introduction

Mitochondria are complex organelles that play important roles in energy metabolism, cell signaling, immune response regulation, apoptosis, ion homeostasis, and other functions (1). Mitochondria contain their own circular double-stranded genome which is approximately 16.5 kilobases in size and encodes 13 proteins, all of which are essential subunits of the electron transport chain, along with two ribosomal RNAs and 22 transfer RNAs (2). Mitochondrial DNA (mtDNA) is packaged into ‘nucleoid’ structures positioned in the inner mitochondrial membrane (3, 4). The mitochondrial nucleoid is composed of over 50 nucleoid-associated proteins involved in maintaining mtDNA and regulating replication and expression (5–8). The most abundant protein present in the nucleoid is Transcription Factor A, Mitochondrial (TFAM). TFAM alone has been shown to be sufficient *in vitro* to condense and compact mtDNA (9, 10). Within a single mitochondrion, multiple copies of mtDNA exist (11, 12) in differing conformations with regard to their nucleoid structure (13). Notably, increased compaction is associated with decreased replication and gene expression activity (14).

mtDNA is more susceptible than nuclear DNA to many damaging agents (15, 16), and it lacks some DNA repair pathways that protect the nuclear genome (15–20). Oxidative damage to mitochondrial genomes has been extensively studied because mitochondria are a major source of production of reactive oxygen species, but oxidative mtDNA damage is efficiently repaired by a robust base excision repair (BER) process. In contrast, the nucleotide excision repair (NER) pathway is absent in mitochondria (20). The absence of NER is particularly salient, because it is the predominant DNA repair pathway that can repair damage caused by a wide range of very common environmental pollutants, including polycyclic aromatic hydrocarbons (PAHs), mycotoxins such as aflatoxin B<sub>1</sub>, aromatic amines, ultraviolet (UV) light exposure, as well as some drugs such as cisplatin (21). The process of mitophagy, the selective degradation of mitochondria, contributes to the removal of irreparable mtDNA damage in human cells (22) and *Caenorhabditis elegans* (23–25). Strikingly, though, despite often incurring high levels of NER-specific forms of DNA damage, the mitochondrial genome appears to be highly resistant to mutagenesis from these lesions. Valente *et al.* (26) did not detect mtDNA mutations in somatic cells of mice exposed to the PAH benzo[a]pyrene or the alkylating agent N-ethyl-N-nitrosourea, despite using exposures that caused nuclear DNA mutagenesis and high levels of mtDNA damage. We reported a similar result in the germline of the nematode *C. elegans* after 50 generations of continuous exposure to the DNA mutagens aflatoxin B<sub>1</sub> or cadmium, even in mitophagy deficient mutants (27). The lack of mutagenesis in these mitophagy deficient mutants suggest that an unknown cellular mechanism prevents mtDNA mutagenesis.

TFAM is a potential candidate for mutagenesis prevention, because it is known to compact the genome and has been suggested to be a damage sensing protein (28). TFAM is a multifunctional protein that serves as both a transcription factor (29–31) and a core component of packaging mtDNA (9, 10, 32) (Figure S1 [C](#)). TFAM is an unusual transcription factor in that it binds specifically to promoters to initiate transcription but also exhibits high affinity binding throughout the mitochondrial genome, presumably to facilitate mtDNA packaging. TFAM is a high-mobility group (HMG) protein with two HMG binding domains that bind to DNA (30, 33–35). Once bound to the DNA, TFAM can multimerize, which facilitates looping and ultimately the compaction of the DNA. It has previously been proposed that TFAM may serve as a DNA damage sensing protein for the mitochondrial genome (28) as TFAM binds differentially to damaged vs. undamaged DNA in the context of oxidative damage (36, 37), base loss (38), cisplatin adducts (39), bulged DNA (40), and alkylated lesions (41). TFAM also interacts with mtDNA repair processes, potentially playing a role in accelerating mtDNA degradation (42) and modulating base excision repair (37, 43). While most

of the forms of mtDNA damage for which TFAM interactions have been characterized can be repaired by mitochondrial base excision repair, interactions between TFAM and irreparable forms of mtDNA damage are less understood.

To further investigate mitochondrial responses to irreparable DNA damage, we used ultraviolet-C (UVC) irradiation. UVC irradiation is ideal for our experiments because it generates almost exclusively photodimers (44) that are irreparable in the mitochondria, but repaired by NER in the nucleus. Using UVC nearly eliminates the occurrence of oxidative damage to DNA or other macromolecules caused by the reactive oxygen species generated by longer-wavelength UV radiation, which could confound interpretation. We characterized cellular responses to UVC exposure using live-cell imaging and gene expression in HeLa cells. We utilized atomic force microscopy (AFM) and protein-DNA binding chips coupled with biochemical assays to characterize human TFAM binding to UVC-induced lesions and structural changes to the compaction status of nucleoids following UVC-induced DNA damage.

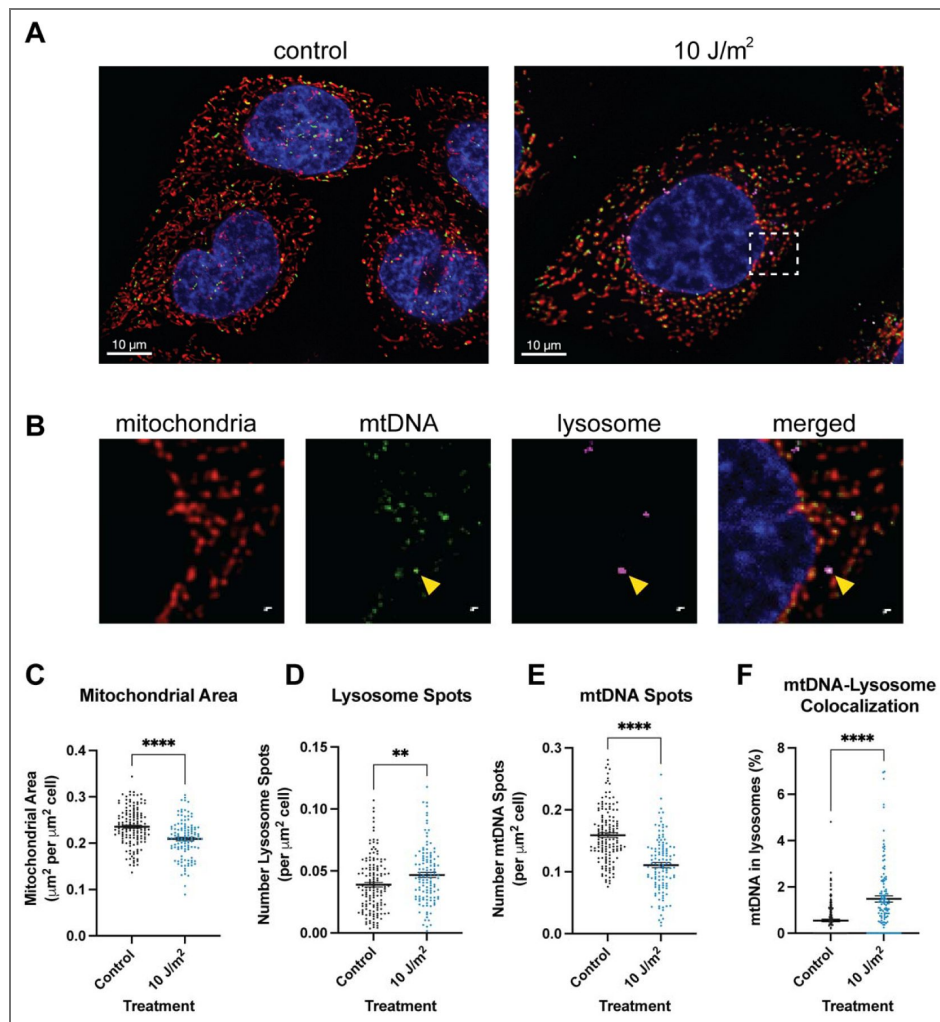
Additionally, we examined the dependence of UVC-induced DNA damage formation and DNA damage levels over time on TFAM protein levels, which has previously been shown to regulate compaction of mitochondrial nucleoids. Overall, our data show that cells respond to irreparable mtDNA damage by increasing mRNA of TFAM and other proteins involved in stimulating mtDNA turnover, in the absence of detectable loss of mitochondrial membrane potential or ATP. *In vitro*, we also characterize dramatic changes in TFAM binding to mtDNA sequences in the context of UVC-induced DNA damage, and show that TFAM compacts UVC-damaged DNA more efficiently than undamaged DNA. *In vivo*, we found that increased TFAM does not protect against UVC-induced mtDNA damage nor alter levels of mtDNA damage over time after UVC exposure, but depletion of TFAM does alter mtDNA damage levels at baseline and after UVC exposure.

## Results

### Mitochondria respond to mtDNA damage from UVC exposure

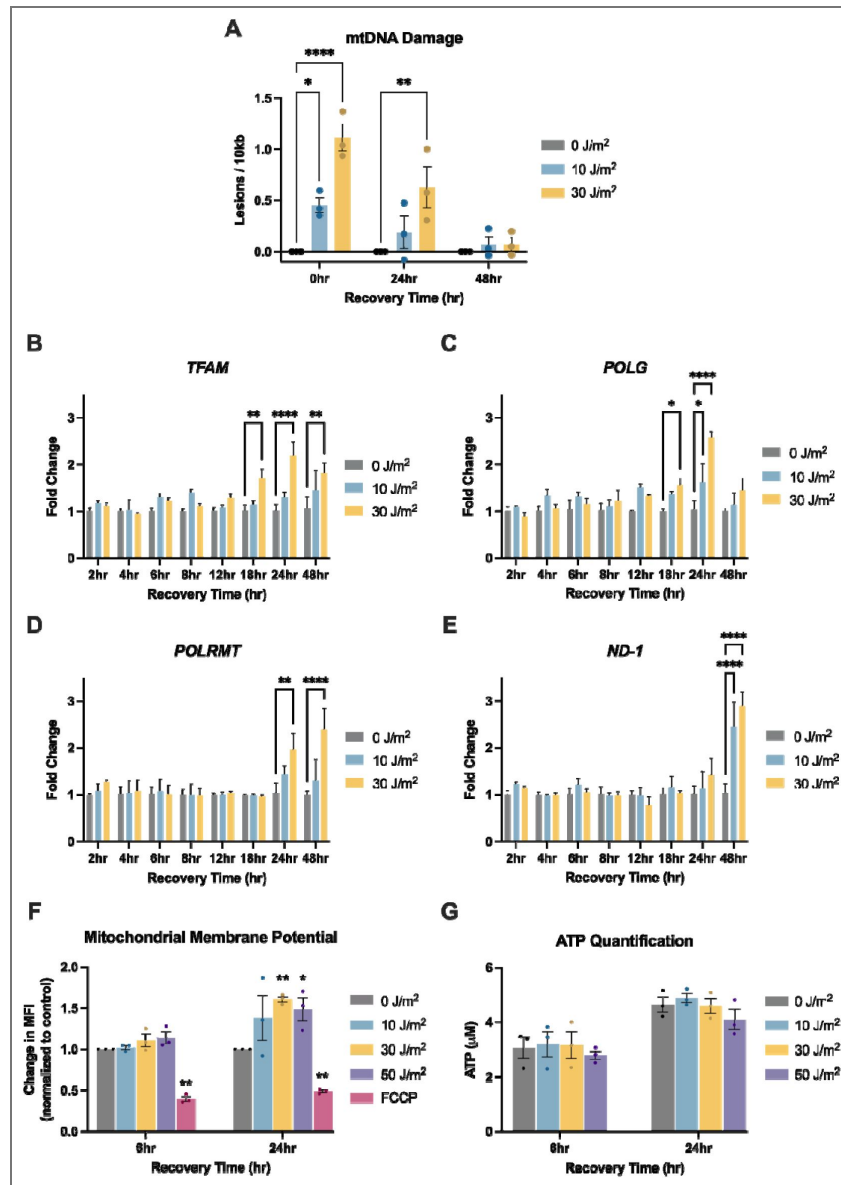
We used multicolor live-cell imaging to investigate alterations to mitochondrial morphology and possible lysosomal trafficking of mtDNA following UVC exposures, after confirmation that these low doses of UVC are not associated with decreases in cell viability (Figure S2 [↗](#)). We performed live-cell imaging on HeLa cells exposed to either 0 or 10 J/m<sup>2</sup> UVC following a 24-hour recovery time after the exposure, staining for mtDNA, lysosomes, mitochondria, and nuclei (Figure 1A, [1B](#) [↗](#)) and quantifying number of mtDNA spots, lysosome spots, area of mitochondria, and colocalization of lysosomes and mtDNA. Because we observe larger cell areas in the UVC treated group (Figure S3A [↗](#)), we normalized our data to the area of each cell. Our data indicate a decrease in the area of mitochondria in UVC treated cells (Figure 1C [↗](#)), but no significant differences in the mitochondrial morphology present in either treatment group, measured by the mean area of individual mitochondria or the mean perimeter (Figure S3B [↗](#), Figure S3C [↗](#)), although there is a modest decrease in mean eccentricity (Figure S3D [↗](#)). Notably, however, the cells exhibit an increase in the total number of lysosomes per cell (Figure 1D [↗](#)) and a decrease in the number of mtDNA spots (Figure 1E [↗](#)). Furthermore, a higher proportion of mtDNA spots colocalize with lysosomes in the UVC treated cells versus the controls (Figure 1F [↗](#), representative image in Figure 1B [↗](#)). Taken together, these results suggest increased mtDNA degradation.

We assessed levels of mtDNA damage immediately following exposure to UVC and after 24 and 48 hours of recovery time. Immediately after damage, we observe a dose-dependent increase in mtDNA lesions with UVC exposure (Figure 2A [↗](#)). After 24 hours, we observe a decrease in mtDNA damage levels, with significant levels of damage still present in cells exposed to 30 J/m<sup>2</sup> (Figure 2A [↗](#)). After 48 hours, damage is below the limit of detection (Figure 2A [↗](#)). The reduction in mtDNA damage could reflect an increase in mtDNA degradation removing the damage, an increase in mtDNA replication helping to dilute the relative amount of damage present in the cells, or a combination of both.



**Figure 1.** Live-cell imaging 24 hours following UVC treatment indicates increased mtDNA degradation.

**A)** Representative images for control (left) and UVC treated (right) cells. Merged channels include mitotracker (red, stains mitochondria), lysotracker (pink, stains lysosomes), picogreen (green, stains mtDNA), and nuclei (blue). **B)** Inset outlined in the UVC treated cells in panel A. Channels from left to right include mitochondria, mtDNA, lysosome, and the merged image. The yellow arrow indicates a mtDNA spot colocalized with lysotracker. For panels C-F, x-axes represent the treatment and data was analyzed via two-tailed unpaired t-test. Data includes at least n=30 cells per treatment group per imaging session and three distinct imaging sessions. **C)** Quantification of the total mitochondrial area per cell normalized to the size of the cell ( $p < 0.0001$ ). **D)** Quantification of the number of lysosomes normalized to the size of the cell ( $p = 0.0054$ ). **E)** Quantification of the number of mtDNA spots normalized to the size of the cell ( $p < 0.0001$ ). **F)** Quantification of the proportion of mtDNA spots within lysosomes ( $p < 0.0001$ ).



**Figure 2.** UVC causes mtDNA damage that decreases over time and is associated with upregulation of mtDNA replication genes, in the absence of apparent mitochondrial dysfunction.

For panels A-F, x-axes represent recovery time, i.e., time following the UVC exposure. Doses of UVC used were 0, 10, and 30 J/m<sup>2</sup>. **A**) Mitochondrial DNA damage levels following UVC exposure. Y-axis represents the level of damage (lesions/10kb). Data analyzed via two-way ANOVA with a Dunnett's post-hoc test for multiple comparisons (dose:  $p < 0.0001$ , time:  $p = 0.0001$ , interaction:  $p = 0.0024$ ). For panels B-E, y-axes represent fold change normalized to the control (0 J/m<sup>2</sup>) at each timepoint. All data was analyzed via two-way ANOVA with Dunnett's post-hoc test for multiple comparisons. **B**) *TFAM* expression level assessed via qPCR following UVC exposure (dose:  $p = 0.003$ , time:  $p < 0.0001$ , interaction:  $p = 0.01$ ). **C**) *POLG* expression level assessed via qPCR following UVC exposure (dose:  $p < 0.0001$ , time:  $p < 0.0001$ , interaction:  $p = 0.001$ ). **D**) *POLRMT* expression level assessed via qPCR following UVC exposure (Dose:  $p = 0.002$ , time:  $p = 0.008$ , interaction:  $p = 0.05$ ). **E**) *ND-1* expression level assessed via qPCR following UVC exposure (Dose:  $p < 0.0001$ , time:  $p = 0.01$ , interaction:  $p = 0.002$ ). **F**) Mitochondrial membrane potential assessed via flow cytometry following exposure to 0, 10, 30, or 50 J/m<sup>2</sup> UVC at 6 and 24 hours after exposure. The x-axis represents the exposure group and timepoint, and the y-axis represents the change in Median Fluorescent Intensities (MFI) of TMRM normalized to the control for each timepoint. Cells were also exposed to FCCP, a well-known chemical that causes mitochondrial depolarization, as a positive control. Data was analyzed via a two-way ANOVA (treatment:  $p = 0.0008$ , time:  $p < 0.0001$ , interaction:  $p = 0.14$ ). **G**) Cellular ATP levels following exposure to 0, 10, 30, or 50 J/m<sup>2</sup> UVC at 6 and 24 hours after exposure. The x-axis represents the exposure group and timepoint, and the y-axis represents the ATP content (μM). Data was analyzed via a two-way ANOVA (treatment:  $p = 0.38$ , time:  $p < 0.001$ , interaction:  $p = 0.94$ ).



To test for a transcriptional program supporting replication activity following UVC exposure, we assessed mRNA levels of key genes (*TFAM*, *PGC1 $\alpha$* , *NRF-1*, *POLRMT*, *POLG*, *ND-1*) associated with mtDNA replication and transcription. *TFAM* is responsible for initiating transcription, and it also plays a role in replication, because mtDNA transcription is coupled to mtDNA replication (45). *TFAM* expression is regulated by *NRF-1*, which is regulated by *PGC1 $\alpha$*  (46). *POLRMT* is the mitochondrial RNA polymerase, which is necessary for transcription as well as replication as it generates primers to initiate mtDNA replication (45). *POLG* is the catalytic subunit of the DNA polymerase responsible for DNA synthesis during mtDNA replication (47). *ND-1* is a transcript of the mitochondrial genome that we utilized to assess mtDNA transcriptional activity.

We observe downregulation of *PGC1 $\alpha$*  and *NRF-1* transcripts starting at 4 hours after exposure to UVC until 24 hours (Figure S4A [↗](#), Figure S4B [↗](#)), at which point *PGC1 $\alpha$*  transcripts increase (Figure S4A [↗](#)), followed by elevation of *NRF-1* transcripts at 48 hours (Figure S4B [↗](#)). Interestingly, downregulation of *PGC1 $\alpha$*  and *NRF-1* at the earlier time points is not associated with downregulation of *TFAM* (Figure 2B [↗](#)). We observe upregulation of *TFAM* (Figure 2B [↗](#)) and *POLG* (Figure 2C [↗](#)) beginning as early as 18 hours after the exposure to UVC. At 24 hours after exposure, we observe continued upregulation of *TFAM* (Figure 2B [↗](#)) and *POLG* (Figure 2C [↗](#)), as well as *POLRMT* (Figure 2D [↗](#)), but not *ND-1* (Figure 2E [↗](#)). These results suggest that within the first 24 hours after the UVC exposure, the cells' nuclear transcriptional program supports increased mtDNA replication, which could resynthesize non-damaged copies of the genome. After a 48-hour recovery period, *TFAM* (Figure 2B [↗](#)), *POLRMT* (Figure 2D [↗](#)) and *ND-1* are upregulated (Figure 2E [↗](#)), suggesting increased transcriptional activity. Taken together, the live-cell imaging results, quantification of mtDNA damage, and gene expression assays demonstrate a cellular response to increase mtDNA turnover that would support both removal and replacement of mtDNA.

A loss of mitochondrial membrane potential is an indirect mechanism that often allows for detection of mitochondrial dysfunction and induction of mitophagy; however, we did not detect a loss of membrane potential in cells exposed to UVC (Figure 2F [↗](#)). In fact, we observed a modest increase in membrane potential following 24 hours after the exposure (Figure 2F [↗](#)), which would not trigger mitophagy. We also tested whether the UVC exposure altered ATP levels, another potential indicator of mitochondrial dysfunction, but saw no differences (Figure 2G [↗](#)). We previously reported a lack of UVC-induced changes in mitochondrial reactive oxygen species levels in primary human fibroblasts (22). Taken together, these results suggest that the mtDNA turnover following damage does not result from indirect sensing of generalized mitochondrial dysfunction and further suggests that there may be a direct damage sensing mechanism.

## High-throughput in vitro data reveals that UVC-induced damage alters TFAM binding specificity across the mitochondrial genome

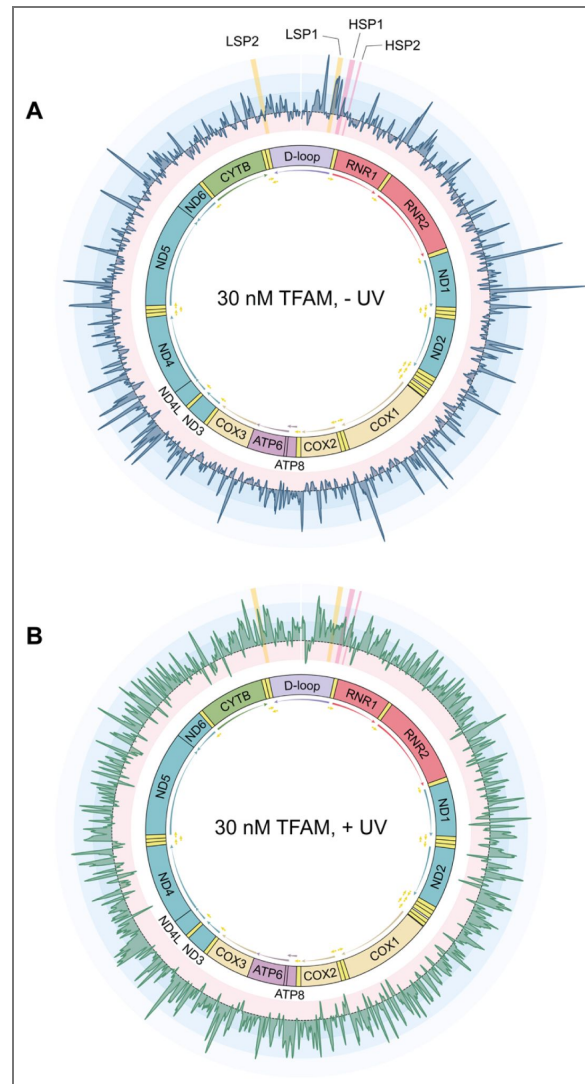
The preceding results suggest that cells might directly sense mtDNA damage. The observations that *TFAM* can compact DNA *in vitro* (9, 10) and exhibits differential binding to damaged DNA (28) suggest that it could potentially serve as a damage sensing protein. In addition, *TFAM* plays a role in maintaining mtDNA copy number and regulating both transcription and replication (29–31, 48). These functions may allow *TFAM* to either “tag” genomes for degradation or sequester them from transcription and replication proteins. We hypothesized that *TFAM* would bind differentially to UVC-induced mtDNA damage, which would be required for such functions. However, in general, the specificity of *TFAM* binding across the mitochondrial genome is not well understood. There are conflicting reports on the specificity with which *TFAM* binds different regions of the mitochondrial genome, with some indicating specificity for promoter regions (49, 50), and others reporting relatively uniform binding across the genome (51, 52). To address these gaps in knowledge and be able to compare *TFAM* binding to damaged vs. undamaged mtDNA, we leveraged an *in vitro* DNA-chip based technology (53) to measure *TFAM* binding to tens of thousands of short DNA sites simultaneously.

We designed a custom library to cover the entirety of the human mitochondrial genome using a sliding 33-nucleotide window width with a 2-nucleotide shift for each consecutive probe (Figure S5). A subset of the chambers was irradiated with UVC to induce photolesions on the DNAs. We then incubated the chambers with either 30 nM or 300 nM TFAM (Figure S5). The concentrations of TFAM were selected based on preliminary experiments to ensure fluorescence intensity values were well within the range of detection. We were able to identify locations in the mitochondrial genome at which TFAM exhibits high occupancy and observed changes in sequence specificity of TFAM binding in the context of UVC-irradiated DNA.

To assess sequence specificity, we normalized TFAM binding levels by converting the fluorescence intensity values to z-scores using a control distribution based on 116 non-mitochondrial low affinity sequences previously identified in a TFAM universal protein-DNA binding array experiment (Methods, Figure S6). Our data reveal high-occupancy TFAM binding sites within the mitochondrial genome that are preferentially bound at the low concentration (30 nM) of TFAM, as indicated by the peaks in the z-scores plotted across the mitochondrial genome (Figure 3). At the high TFAM concentration (300 nM), peaks in binding occupancy occur at the same locations in the genome; however, their z-scores are lower due to the overall increase in TFAM binding to all sequences (Figures S7 and S8). We observe a strong correlation in z-scores between probes in the 30 nM and 300 nM datasets, further supporting the specificity of TFAM binding even at a high TFAM concentration (Figure S8). Previously, it has been suggested that the minimal binding motif for TFAM is two guanines, separated by 10 random nucleotides, referred to as a GN<sub>10</sub>G motif (35) because all four TFAM promoter sequences contain this motif and crystal structures indicate interactions between TFAM and the guanine nucleotides in this motif (35, 54). Our data, however, does not support this hypothesis. Specifically, the distributions of z-scores for sequences with and without the GN<sub>10</sub>G motif are strikingly similar, with slightly lower levels of TFAM occupancy indicated at sequences containing these motifs (Figure S9). Comparison of our data with previously published *in vivo* mitochondrial DNase I footprints (49) show strong overlap between the two data sets (Figure S10), suggesting that these data may be representative of *in vivo* TFAM binding. Additional comparison of our data with recent *in vitro* TFAM footprinting performed on linear mtDNA using Fiber-seq (13) indicates strong agreement between the two methods (Figure S11).

mtDNA contains four promoters that are regulated by TFAM: two on the heavy strand (HSP1 and HSP2) and two on the light strand (LSP1 and LSP2). Within promoter regions of the mitochondrial genome, we observe peaks in z-scores at both LSP1 and HSP1, but not LSP2 and HSP2 promoters (Figure 3, Figure S12). However, the peak in HSP1 is located in the transcription start site rather than the TFAM binding site (Figure S12). Interestingly, the z-scores at these sites are lower than many of the high-occupancy sites identified throughout the genome (Figure 3A). Although perhaps surprising at first glance, previous work also indicates a lack of enrichment of TFAM at promoter sequences *in vivo* (51), and modest footprints of TFAM binding at promoter sites *in vitro* (54).

In contrast to the large peaks in the z-scores that we observe with non-irradiated DNA, experiments performed on irradiated DNA show more uniform binding patterns and reduction in size of distinct peaks in the z-scores throughout the mitochondrial genome (Figure 3B). These results suggest that the UVC damage reduces sequence specificity of TFAM. To examine this finding more systematically, we plotted the distribution of z-scores associated with the probes that are in the top 5% of z-scores and the bottom 5% of z-scores in the context of non-damaged DNA and UVC-irradiated DNA (Figure S13A-B). The z-scores of our top binders (*i.e.*, probes for which TFAM exhibits high specificity) decrease after UVC irradiation (Figure S13A), whereas the z-scores of our weakest binding sequences (*i.e.*, probes for which TFAM exhibits little to no preference) increase after UVC irradiation (Figure S13B). In addition, plotting the median z-score in the non-UVC-irradiated chamber against the change in z-score after UVC-irradiation reveals a strong negative correlation, further supporting a reduction in sequence specificity upon UVC irradiation (Figure S13C).



**Figure 3. TFAM has specific binding across the mitochondrial genome and exhibits a reduction in specificity in the context of UVC-irradiated DNA.**

The median z-score plotted to the coordinate of the middle nucleotide of the variable mitochondrial region of the sequence for the non-UVC-irradiated chamber containing 30 nM TFAM (**A**) and the chamber irradiated with 1,500 J/m<sup>2</sup> UVC containing 30 nM TFAM (**B**). The gene map of the mitochondrial genome is shown in the center. Z-score variation is color coded such that positive z-scores associated with high binding are in blue, and progressively get lighter as the z-scores get higher. Negative z-scores associated with low binding are in red. Regions highlighted in yellow are the promoter sequences of the mitochondrial genome on the light strand (LSP1 and LSP2), while regions highlighted in pink are the promoter sequences on the heavy strand (HSP1 and HSP2). Plots of these regions can be found in [Figure S12](#).



## TFAM has high affinity for DNA and undergoes cooperative binding regardless of UVC-induced damage

The high-throughput binding assays reveal that TFAM exhibits DNA sequence preferences, but these preferences cannot be directly extrapolated to the binding affinity of TFAM to DNA, nor can they provide insight on the stoichiometry of binding. To investigate differences in TFAM binding affinity that may be associated with UVC-induced DNA damage, we performed fluorescence anisotropy and fit the binding curve to the Hill Equation (55) to determine the  $K_D$  and the extent of cooperative binding ( $n$ ) (Methods). Sequences were selected to cover a range of z-scores identified in our array-based binding assays, and anisotropy experiments were performed on both undamaged and UVC-irradiated oligonucleotides. All sequences analyzed exhibit  $K_D$  values between 3 nM and 9 nM and  $n$  values between 1 and 5, indicating multiple TFAM proteins binding to each DNA. Overall, we do not observe any large differences in  $K_D$  or cooperativity values with or without UVC irradiation across any of the sequences analyzed (Figure S14, Table 1). This lack of correlation between the array-based binding data and the direct measurements of binding affinity contrasts with previous studies in which the occupancy of proteins at different DNA sequences, measured on high-density DNA arrays, directly correlates with the binding affinity (53, 56, 57), and suggests that the differences in TFAM occupancy on the arrays may result from differences in extents of cooperative binding that is not discernable in the fluorescence anisotropy experiments.

To further examine the extent of oligomerization of TFAM on these DNA oligonucleotides and determine if there are any differences in TFAM binding on the “high” and “low” occupancy sequences (selected according to the array-based binding occupancy measurements), we used atomic force microscopy (AFM) to examine the oligomeric state of TFAM in the absence and presence of one of the high and one of the low occupancy sequences (Table 1). AFM is a powerful tool for determining the multimeric state of proteins, because the AFM volume correlates linearly with protein molecular weights (58, 59) and AFM can be used to identify cooperative binding on small DNA oligonucleotides (60). Inspection of the volume distributions of TFAM in the absence of DNA reveals a single peak at  $\sim 35 \text{ nm}^3$  with a tail to higher volumes, consistent with the dominant species being a monomer and the existence of a small amount of higher order species (Methods, Figure S15A). In the presence of the oligonucleotides, the volume distribution shifts to higher volumes that are consistent with dimers and higher order species (Figure S15B, C). Notably the “high occupancy sequence” shows two distinct peaks at  $\sim 50 \text{ nm}^3$  and  $110 \text{ nm}^3$ , consistent with cooperative assembly of a dimer and tetramer of TFAM on this oligonucleotide. In contrast, the “low occupancy sequence” exhibits a broad distribution with a peak at  $\sim 50 \text{ nm}^3$  and a long tail to higher volumes, consistent with the cooperative formation of a dimer followed by weaker assembly of higher order species. This is consistent with previous SEC-MALS studies showing that TFAM oligomerization can differ with varying sequences (61). These observations in differences in binding properties may not be captured in the fluorescence anisotropy measurement and therefore may account for the lack of correlation between the high-throughput array-based binding data and the binding affinities measured by fluorescence anisotropy.

## TFAM compacts UVC-damaged DNA more efficiently than undamaged DNA

The array-based TFAM binding data indicate that the relative occupancy of TFAM for sequences in the mitochondrial genome change upon irradiation, with lower specificity associated with irradiation. Because one of the roles of TFAM is to compact the mitochondrial genome into nucleoid structures (9, 10), we examined the impact of UVC-induced DNA damage on nucleoid structure using AFM. For these experiments, we used circular plasmid DNA (pUC19) that was either untreated or exposed to  $100 \text{ J/m}^2$  UVC, which introduces roughly three photolesions per plasmid (Methods). We incubated the damaged or undamaged DNA with TFAM for two minutes

**Table 1.** Binding of TFAM to array sequences, measured using fluorescence anisotropy

| Sequence      | -UV           |                           |                    | +UV           |                           |                    |
|---------------|---------------|---------------------------|--------------------|---------------|---------------------------|--------------------|
|               | Array z-score | $K_D \pm \text{SEM}$ (nM) | $n \pm \text{SEM}$ | Array z-score | $K_D \pm \text{SEM}$ (nM) | $n \pm \text{SEM}$ |
|               | percentile    |                           |                    | percentile    |                           |                    |
| ND4_473*      | 0.48          | $4.95 \pm 0.03$           | $5.06 \pm 0.09$    | 28.28         | $5.36 \pm 0.14$           | $4.76 \pm 0.11$    |
| ND2_401       | 25.32         | $4.36 \pm 0.24$           | $2.64 \pm 0.22$    | 23.62         | $6.24 \pm 0.15$           | $2.61 \pm 0.25$    |
| ND3_92        | 25.68         | $4.53 \pm 0.13$           | $2.63 \pm 0.22$    | 10.23         | $5.54 \pm 0.14$           | $2.19 \pm 0.06$    |
| RNR2_619      | 25.71         | $4.89 \pm 0.33$           | $2.46 \pm 0.23$    | 51.19         | $8.13 \pm 0.39$           | $2.33 \pm 0.10$    |
| COX1_27       | 75.32         | $6.09 \pm 0.43$           | $2.07 \pm 0.07$    | 74.45         | $5.90 \pm 0.10$           | $2.12 \pm 0.28$    |
| ND1_288       | 75.16         | $3.41 \pm 0.03$           | $3.04 \pm 0.19$    | 35.79         | $6.12 \pm 0.22$           | $1.90 \pm 0.12$    |
| ND6_87        | 75.74         | $5.20 \pm 0.32$           | $2.84 \pm 0.14$    | 95.50         | $7.63 \pm 0.21$           | $2.27 \pm 0.26$    |
| ND1_450       | 90.25         | $4.33 \pm 0.17$           | $2.40 \pm 0.43$    | 70.40         | $6.36 \pm 0.51$           | $2.20 \pm 0.13$    |
| COX2_229      | 90.78         | $3.94 \pm 0.03$           | $2.77 \pm 0.17$    | 80.81         | $4.87 \pm 0.35$           | $2.18 \pm 0.14$    |
| TRNT_10       | 90.90         | $4.24 \pm 0.12$           | $2.15 \pm 0.24$    | 96.07         | $7.79 \pm 0.80$           | $1.81 \pm 0.05$    |
| ND1_353 40mer | 99.99         | $5.69 \pm 0.15$           | $3.09 \pm 0.11$    | 94.24         | $7.25 \pm 0.39$           | $4.38 \pm 0.33$    |
| ND1_353**     | 99.99         | $7.56 \pm 0.34$           | $3.85 \pm 0.49$    | 94.24         | $7.66 \pm 0.88$           | $2.94 \pm 0.40$    |

\* This sequence was used for AFM oligomerization studies to represent a low occupancy sequence in the array-based TFAM binding data, referred to as "low occupancy sequence" in the text

\*\* This sequence was used for AFM oligomerization studies to represent a high occupancy sequence in the array-based TFAM binding data, referred to as "high occupancy sequence" in the text

before depositing the sample on the mica surface. We used low concentrations of TFAM (15 nM or 30 nM) to prevent full compaction of the DNA and allow us to capture the differences, if any, in the extents of compaction.

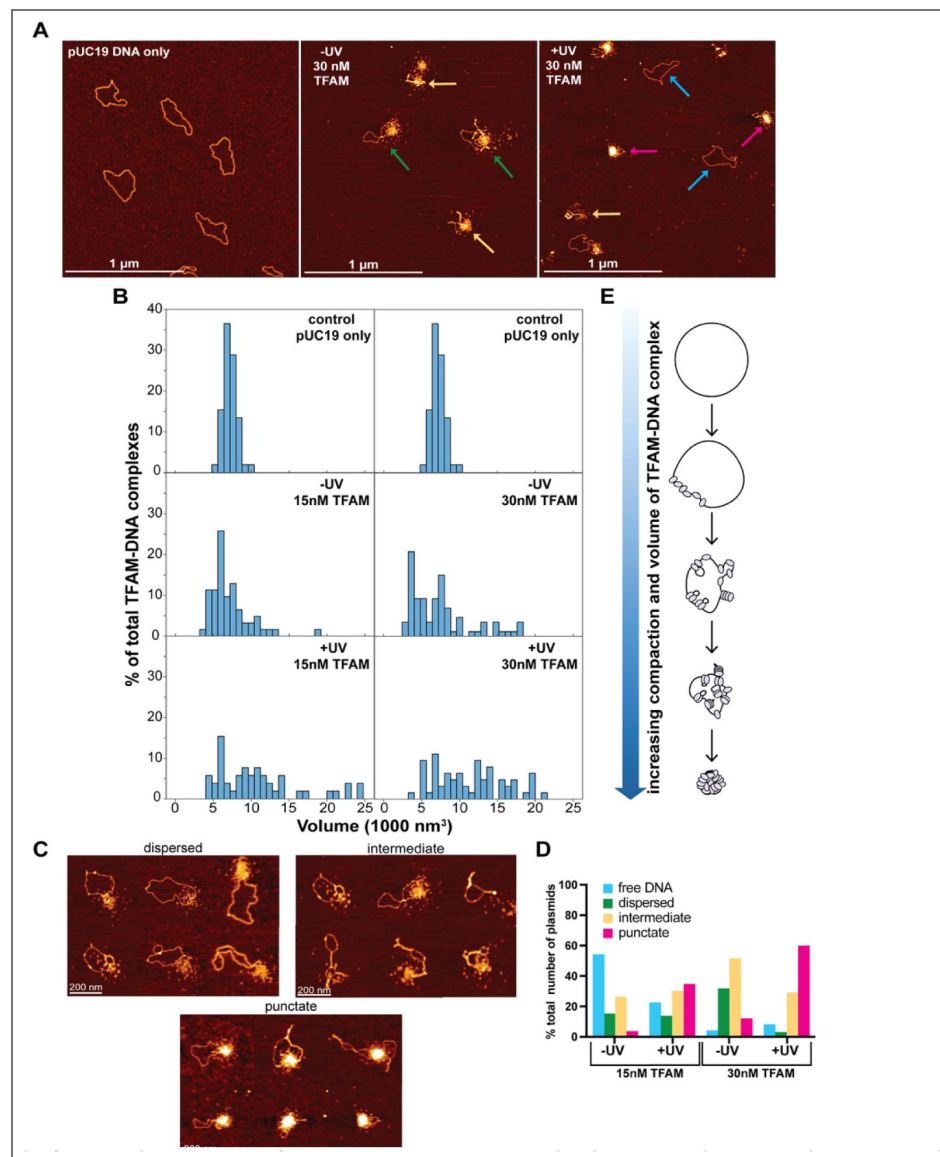
Figure 4A shows images of pUC19 alone as well as undamaged and damaged pUC19 with 30 nM TFAM. The AFM images show that in the absence of TFAM, for both the undamaged pUC19 and the damaged pUC19, the plasmid is well spread on the surface (Figure S16). In contrast, TFAM promotes the formation of both punctate and disperse protein assemblies, as well as tracts of protein along the DNA, which often bridge two regions of DNA together (Figure 4A, S17). Notably, most DNAs show only a single cluster of proteins, with the majority of the rest of the DNA being unbound. In fact, at 15 nM TFAM a significant percent of the DNAs have no protein bound, while other DNA molecules have punctate assemblies (Figure S18). These observations indicate that TFAM is cooperatively assembling on the DNA both in the presence and absence of irradiation, consistent with previous work indicating TFAM is highly cooperative (62).

To assess the extent of compaction, we measured the total volume of the DNA and its associated proteins on DNA (hereafter referred to as “complexes”) (Methods). In the absence of protein, the distribution of volumes shows a single peak centered at  $\sim 5000 \text{ nm}^3$  (Figure 4B). Addition of 15 nM and 30 nM TFAM results in the distributions broadening and shifting to higher volumes with increasing TFAM concentration in both the absence and presence of UVC irradiation. Notably, the volumes for the irradiated DNAs are greater than those without irradiation: the volumes for samples with irradiated DNA at 15 nM and 30 nM TFAM are  $\sim 12,600 \text{ nm}^3$  and  $\sim 12,700 \text{ nm}^3$ , respectively, and the samples without irradiation at 15 nM and 30 nM TFAM are  $\sim 8,250 \text{ nm}^3$  and  $\sim 7,500 \text{ nm}^3$ , respectively (Figure 4B). These results suggest that UVC irradiation increases the cooperative assembly of TFAM on pUC19. Inspection of the AFM images reveals that these differences in volume are associated with differences in compaction of the DNA. Specifically, addition of TFAM to UVC irradiated DNA results in more compact and punctate structures relative to undamaged DNA, which shows proteins that appear to be more loosely associated with the DNA (Figure 4A, 4C, S18).

To quantitatively assess the differences in compaction, we categorized each TFAM-pUC19 complex as: free DNA, dispersed (small clusters with no protein tracts on DNA), intermediate (small clusters with protein tracts), and punctate (tightly associated punctate clusters) (Figure 4C). At 15 nM TFAM, the majority of the DNAs ( $\sim 55\%$ ) have no protein bound with very few punctate species with unirradiated DNA; whereas, the irradiated DNA shows significantly lower percentage of free DNA and an increase of punctate structures (Figure 4D, Table S1). At 30 nM TFAM, most of the DNAs show dispersed ( $\sim 35\%$ ) and intermediate ( $\sim 60\%$ ) species on unirradiated DNA and intermediate ( $\sim 30\%$ ) and punctate ( $\sim 60\%$ ) species on irradiated DNA. Taken together, these results indicate that UVC-induced DNA damage promotes compaction of DNA by TFAM (Figure 4E).

## Increased TFAM levels do not offer protection from UVC-induced mtDNA damage or alter accumulated DNA damage levels over time

Recent evidence shows that nucleoid compaction varies throughout cellular differentiation and between cell types (13). It is well known that chromatin both protects from various forms of DNA damage and regulates DNA repair activities (63, 64); however, little is known about the role of DNA compaction in the mitochondria. Although it has been proposed that the mitochondrial nucleoid may protect mtDNA from damage (14), there is no direct evidence supporting this. Our observation that TFAM compacts damaged DNA suggests that mtDNA compaction may play a role in protecting the cell from damaged mtDNA. To examine these possibilities, we tested whether overexpressing TFAM, which has been shown to increase mtDNA compaction in cells (13), would protect mtDNA from DNA damage or promote the removal of damaged mtDNA. To carry out these



**Figure 4. Atomic force microscopy of TFAM-DNA substrates indicates an increase in compaction associated with UVC exposure.**

**A)** AFM images of the pUC19 DNA only (control) and the 30 nM TFAM-DNA complexes in two different conditions: one with (+UV) or without (-UV) UVC irradiated DNA. Colored arrows on the AFM images represent the different TFAM-DNA complexes categorizations: dispersed (green), intermediate (yellow), punctate (red), and free DNA (blue). The white scale bar represents 1  $\mu\text{m}$ . UV irradiated plasmids were exposed to 100 J/m<sup>2</sup> UVC. **B)** Histogram plot of the volumes distribution of plasmid DNA only (control) as well as the 15 nM and 30 nM TFAM-DNA complexes with (+UV) or without (-UV) UVC damaged DNA. All axes in the histogram are scaled the same. The data for control pUC19 only was replicated for each TFAM concentration for clarity in comparisons. **C)** Atomic force microscopy images of three different TFAM-DNA complex categorizations labeled as dispersed (small clusters with no protein tracts on DNA), intermediate (small clusters with protein tracts), and punctate (tightly associated punctate clusters). The white scale bar represents 200 nm. **D)** A bar graph representing the percent total number of plasmids in the 15nM TFAM concentration with (+UV) (N=66) or without (-UV) (N=136) UVC damaged DNA and the 30nM TFAM concentration with (+UV) (N=65) or without (-UV) (N=91) UVC damaged DNA. Detailed counts of each classification can be found in Table S1. **E)** Schematic of the TFAM-DNA binding and compaction mechanism.

experiments, we used the same TFAM overexpression cell line generated by Isaac *et al.*, which was previously used to show that increasing TFAM level increases the number of inaccessible (presumably compacted) nucleoids (13).

Following a 48-hour pretreatment with doxycycline (schematic in Figure 5A), we observe a two-fold induction of TFAM protein, measured by western blot (Figure 5B), quantified in Figure 5C), as well as an increase in mRNA levels (Figure 5D). Elevated TFAM levels are sustained throughout the duration of the experiment (Figure 5B, Figure 5C, Figure 5D). To test whether increased TFAM expression is associated with protection of mtDNA from UVC-induced DNA damage, we exposed these cells to UVC and quantified the amount of mtDNA damage they received. Across a range of UVC doses, there are no differences in the lesion frequency between the control and TFAM overexpressing cells (Figure 5E). Additionally, overexpression of TFAM does not influence the accumulated DNA damage levels over time, evidenced by the similar lesion frequencies observed 24 hours and 48 hours following the UVC exposure (Figure 5F, Figure 5G), nor does overexpression of TFAM protect the cells from a loss of cell viability following exposure to UVC (Figure S19).





represents the fold change relative to the non-doxycycline treated cells. Data was analyzed via two-way ANOVA (doxycycline treatment  $p=0.0003$ , time  $p=0.52$ , interaction  $p=0.52$ ). **E**) Lesion frequency following UVC exposure in TFAM-tetON cells immediately after the exposure (0-hour recovery time). X-axis represents the doxycycline treatment across a range of UVC doses and y-axis represents the lesion frequency (lesions per 10kb). Data was analyzed via two-way ANOVA (UVC dose:  $p<0.0001$ , doxycycline treatment  $p=0.15$ , interaction:  $p=0.77$ ). **F**) Lesion frequency following UVC exposure in TFAM-tetON cells 24 hours after the exposure (24-hour recovery time). X-axis represents the doxycycline treatment across a range of UVC doses and y-axis represents the lesion frequency (lesions per 10kb). Data was analyzed via two-way ANOVA (UVC dose:  $p<0.0001$ , doxycycline treatment  $p=0.71$ , interaction:  $p=0.65$ ). **G**) Lesion frequency following UVC exposure in TFAM-tetON cells 48 hours after the exposure (48-hour recovery time). X-axis represents the doxycycline treatment across a range of UVC doses and y-axis represents the lesion frequency (lesions per 10kb). Data was analyzed via two-way ANOVA (UVC dose:  $p<0.0001$ , doxycycline treatment  $p=0.01$ , interaction:  $p=0.61$ ). **H**) Representative AFM images of *in vitro* nucleoids generated using purified TFAM and PCR amplified human mtDNA at 0, 100, 250, and 1000 nM TFAM. All scale bars represent 500 nm. **I**) Lesion frequency following UVC exposure in *in vitro* nucleoids. X-axis represents the UVC dose and y-axis represents the lesion frequency (lesions per 10kb). Data was analyzed via two-way ANOVA (UVC dose:  $p<0.0001$ , TFAM concentration  $p=0.20$ , interaction:  $p=0.86$ ). **J**) Lesion frequency following UVC exposure in cells with and without a TFAM knockdown. The x-axis represents the recovery time, i.e., time following exposure to UVC and the y-axis represents the level of mtDNA damage (lesions per 10kb). Data was analyzed via a three-way ANOVA (UVC dose:  $p<0.0001$ , recovery time:  $p=0.002$ , siRNA treatment:  $p<0.0001$ , recovery time\*UVC dose:  $p<0.0001$ , recovery time\*siRNA treatment:  $p=0.001$ , UVC dose\*siRNA treatment:  $p=0.08$ , recovery time\*UVC dose\*siRNA treatment:  $p=0.66$ ). **K**) Ratio of mtDNA copy number to nuclear DNA copy number following UVC exposure in cells with and without a TFAM knockdown. The x-axis represents the recovery time, i.e., time following exposure to UVC and the y-axis represents the level of mtDNA damage (lesions per 10kb). Data was analyzed via a three-way ANOVA (UVC dose:  $p=0.01$ , recovery time:  $p<0.01$ , siRNA treatment:  $p<0.0001$ , recovery time\*UVC dose:  $p<0.001$ , recovery time\*siRNA treatment:  $p<0.0001$ , UVC dose\*siRNA treatment:  $p=0.04$ , recovery time\*UVC dose\*siRNA treatment:  $p=0.01$ ).

To test if the level of compaction of the mitochondrial nucleoid is protective against mtDNA damage across a wider range of TFAM concentrations, we generated *in vitro* reconstituted nucleoids with full-length human mtDNA and varying amounts of TFAM. We generated a series of dose-response curves for UVC to determine if elevated TFAM levels are associated with protection from UVC-induced DNA damage. The conformational status of the mtDNA nucleoids was confirmed using AFM (Figure 5H). We observe that increased nucleoid compaction, driven by elevated TFAM concentrations, is not sufficient to protect mtDNA from UVC-induced lesions (Figure 5I). Taken together, these results indicate that the degree of TFAM mediated compaction of mtDNA does not affect the amount of DNA damage from UVC, and that a roughly two-fold increase in TFAM protein does not alter the accumulated mtDNA damage levels over time. Finally, we tested whether knockdown of TFAM would alter mtDNA damage levels in HeLa cells. We found that TFAM knockdown progressively increased mtDNA damage over time in the absence of UVC exposure as previously reported (37), and resulted in a persistence of mtDNA damage over time after UVC exposure (Figure 5J), likely in part due to the reduced mtDNA copy number and replication capacity following the knockdown of TFAM (Figure 5K).

## Discussion

The mechanisms by which cells handle mtDNA damage remain poorly understood. In particular, the absence of NER in mitochondria begs the question of how bulky lesions and photodimers in mtDNA are removed. TFAM is a multifunctional protein that regulates both transcription and replication of mtDNA, and it compacts the genome into nucleoids. Previously, other groups have hypothesized that TFAM may protect the DNA from DNA damage (14) and serve as a damage sensing protein (28). In this work, we investigate cellular responses to low levels of mtDNA damage caused by UVC irradiation as well as the effect of UVC damage on TFAM-DNA interactions. The results from this study indicate that although TFAM does not protect against UVC-induced DNA damage, it does alter mtDNA compaction *in vitro*, alters mtDNA damage levels with and without

UVC exposures *in vivo*, and suggest that it may function as a photolesion DNA damage sensing protein. This work represents the first step towards understanding how TFAM may interact with irreparable NER-specific substrates to prevent mutagenesis in the mitochondrial genome.

## Mitochondria respond to UVC-induced mtDNA damage in the absence of apparent mitochondrial dysfunction

Exposure to UVC that induces mtDNA damage results in a host of responses, including removal of mtDNA and an increase in gene expression of key mtDNA replication genes, presumably to replenish healthy copies of mtDNA (Figure 1 [↗](#), 2 [↗](#)). The removal of mtDNA begins within the first 24 hours after damage exposure, indicated by an increase in mtDNA present within lysosomes as well as a decrease in the number of mtDNA spots present in live-cell imaging experiments (Figure 1 [↗](#)). This removal of mtDNA occurs alongside upregulation of *TFAM*, *POLG*, and *POLRMT*, all of which are involved in mtDNA replication (Figure 2 [↗](#)). The observed decrease in mtDNA damage levels over time after UVC exposures (Figure 2A [↗](#)) could be a result of both the removal of mtDNA and an increase in the synthesis of non-damaged mtDNA molecules; additional work will be required to distinguish these possibilities. 48 hours after the exposure to UVC, cells upregulate *TFAM* and *POLRMT* as well as transcriptional activity on mtDNA (measured by upregulation of *ND-1*) (Figure 2 [↗](#)), likely to replenish mtRNA and protein levels.

These damage-induced changes to mtDNA transcription and the transcription of nuclear-encoded genes involved with mtDNA replication (*TFAM*, *POLG*, *POLRMT*) occurred in the absence of apparent mitochondrial dysfunction, indicated by no detectable changes to mitochondrial membrane potential or ATP levels (Figure 2F, G [↗](#)). Previously, we have also shown that UVC exposures do not result in a detectable increase in mitochondrial reactive oxygen species (mtROS) (22), though it should be noted that this study used a different type of cell and included a thymidine block. While other possible manifestations of mitochondrial dysfunction after UVC exposure still cannot be ruled out, the current absence of any detectable indications of mitochondrial dysfunction following UVC irradiation suggest the possibility that cells might directly detect UVC-induced mtDNA damage.

## TFAM exhibits preferential binding to a large number of sequences in the mitochondrial genome

Examination of the array-based binding data reveal that in the absence of damage, TFAM shows a wide range of z-scores across the mitochondrial genomic sequences (Figure 3 [↗](#)), indicating preference of certain sequences. These data are in agreement with previously published DNase footprinting data (49) (Figure S10 [↗](#)) and the recently published Fiber-seq footprinting technique used to map TFAM binding patterns to linear mtDNA (13) (Figure S11 [↗](#)). In contrast, previously published ChIP-seq data for TFAM does not show specific enrichment, i.e. no significant peaks (51, 65). However, given the heterogeneity of compaction of mitochondrial genomes in cells and the fact that many genomes in each cell are fully coated with TFAM (13, 14), it is not surprising that ChIP assays do not result in strong TFAM enrichment patterns. Interestingly, the sequences in our data with high z-scores are not associated with the GN<sub>10</sub>G motif, which was previously suggested as a consensus sequence for TFAM (Figure S9 [↗](#)) (35). However, this is not surprising given that the higher affinity of TFAM for this motif in the original publication (35) was only observable at high salt concentrations but not at physiologic salt concentrations like those utilized in our study. This is consistent with Fiber-seq analysis of TFAM binding which also failed to observe significant enrichment at GN<sub>10</sub>G motifs in the mitochondrial genome (13). In addition, we found that the z-scores at promoter sites were modest compared to other regions in the genome (Figure 3 [↗](#) and Figure S12 [↗](#)). These findings indicate that unlike most transcription factors, TFAM by itself shows low specificity for its promoters, with the exception of LSP1, and suggest that TFAM may require interactions with additional proteins present in the nucleoid to promote transcription. Indeed, both earlier work from Gaspari *et al.* (66) and recent work from Tan *et al.* (Figure S4a [↗](#)) (54) indicates that POLRMT and TFB2M are also needed, in addition to TFAM, to produce a notable enrichment of TFAM at the promoters.

Although we do not see strong binding signal at the promoter sites other than LSP1, the preferential binding of TFAM to sequences across the genome may be involved with facilitating compaction of the genome, which could in turn regulate transcription and replication. Specifically, the extent of the compaction of the mitochondrial genome is inversely proportional to the transcription and replication activity (14), similar to nucleosomes.

While it is clear from our array data and other studies (13, 49) that TFAM exhibits sequence specificity, no specific binding motifs have been identified. The sequences we determined to have the highest TFAM binding levels on our array do not overlap with known regulatory or structural elements within the mitochondrial genome, such as origins of replication, promoter sites, or secondary structures that may form on the arrays. There is however, one exception to this in the D-loop where we observe a peak in TFAM binding level upstream of LSP1 at the conserved sequence block 2 (CSB2) (Figure 3A). Interestingly, there is a discrepancy between our TFAM binding array results and our determined  $K_D$  values for these same sequences via fluorescence anisotropy (Table 1) in that sequences for which we observe stronger TFAM signal on the arrays do not have tighter binding affinity. This suggests to us that the differences in TFAM binding levels that we observe on the arrays may be driven by another feature, such as differences in TFAM cooperativity. Indeed, using AFM, we determined that for a sequence for which we observe higher binding levels on the array, TFAM undergoes increased oligomerization compared to a sequence for which we observed lower TFAM binding levels on the array (Figure S15). It remains unknown what about these specific sequences drive differences in TFAM cooperativity. Given that TFAM is known to bend DNA in order to facilitate compaction (67) and has been proposed to induce small ssDNA bubbles (62), one possibility is that differences in either DNA flexibility or local melting abilities could be influencing TFAM's sequence specificity, but this remains to be tested.

## TFAM does not protect DNA from UVC-induced damage but more readily compacts irradiated DNA

Previous studies have indicated that TFAM promotes DNA compaction (9, 10). Examination of complexes of TFAM bound to pUC19 and linear full length human mtDNA using AFM show that TFAM promotes compaction of both DNAs, with the extent of compaction dependent on TFAM concentration (Figure 4A, S18, 5H). At low TFAM concentrations, we observe small punctate complexes with tracts along the DNA (Figure S17) which convert into large punctate complexes at higher TFAM concentrations (Figure 4A).

In the nuclear genome, compaction of DNA serves to protect DNA from damage as well as regulate expression, replication, and repair of DNA. Mitochondrial nucleoid compaction is believed to be associated with regulatory properties such as expression and replication (14). TFAM-induced DNA compaction has been shown to protect mtDNA from enzymatic processes used in techniques such as Fiber-seq (13); however, it is unknown if TFAM protects DNA from damage. To determine if nucleoid compaction protects DNA from UVC-induced damage, we utilized both *in vitro* reconstituted nucleoids and an inducible TFAM overexpression cell model to investigate whether the compactional state of mitochondrial nucleoids serves to protect the DNA from damage. *In vitro*, even at the highest concentration of TFAM, where the entire mtDNA appears compact in AFM images, we observe no protection of DNA from UVC (Figure 5H, Figure 5I). Similarly, in cells where TFAM is overexpressed ~2-fold, which others have shown is associated with an increase in the number of inaccessible (presumably compacted) nucleoids (13), we observe no protection from UVC-induced DNA damage or changes in accumulated DNA damage levels (Figure 5E, F, G). Taken together, these results indicate that TFAM does not appear to protect DNA from UVC-induced DNA damage formation. However, it remains possible that increased compaction of mtDNA by TFAM could protect DNA from other types of DNA damage that form as a result of chemical or enzymatic processes such as reactive oxygen species or alkylating agents that require the DNA to be accessible. This may be particularly true for larger DNA-reactive chemicals such as PAHs or mycotoxins. Finally, while TFAM overexpression in cells does not alter the levels of mtDNA damage after UVC over time, TFAM knockdown is associated with an increase in baseline

mtDNA damage in cells over time (in the absence of UVC), and also results in a persistence in mtDNA damage over time after UVC exposure (Figure 5J). Additional work will be required to discriminate whether the persistence in mtDNA damage after UVC exposure in the TFAM knockdown is because of slower mtDNA removal, slower dilution via reduced mtDNA replication, increased damage by endogenous processes, or some other factor.

Despite not protecting mtDNA from UVC-induced DNA damage, our AFM and array-based binding data suggest that TFAM may act as a damage sensing protein via increased binding and its compaction-promoting function. Our AFM data show significantly increased TFAM-mediated compaction of irradiated vs unirradiated pUC19 DNA (Figure 4). Specifically, in the presence of irradiation, TFAM promotes compact nucleoid structures, whereas in the absence of irradiation, the TFAM clusters are more loosely associated (Figure 4, S18). These results suggest that UVC irradiation promotes the cooperative assembly of TFAM on DNA. These TFAM assemblies do not appear to supercoil the DNA, but rather appear to bring distal DNA segments together, consistent with previous studies (Figure 4, S17) (9). As such, UVC lesions could potentially serve as nucleation sites for TFAM assemblies, and these local clusters may come together to form a compact nucleoid structure containing large regions of the DNA. This suggestion is consistent with laser tweezer studies on stretched DNA that showed formation of multiple small linear clusters of TFAM that can come together to form a larger linear cluster (62). In our studies, the DNA is not stretched and therefore, the TFAM assemblies can come together through space to form compact nucleoids formed via TFAM-DNA interactions that promote further oligomerization of TFAM with itself. Notably, UVC appears to significantly stabilize TFAM-TFAM interactions to promote a compact nucleoid. This increased compaction of DNA by TFAM following UVC irradiation represents a novel role for TFAM which may enable TFAM to change the architecture of mitochondrial genomes harboring DNA damage. The degree of compaction of mtDNA *in vivo* following DNA damage remains to be tested and will need be assessed on a single-nucleoid level given the already heterogeneous nature of mtDNA packaging.

Our high-throughput TFAM binding data reveal a redistribution of z-scores upon UVC irradiation, with more uniform distributions of TFAM occupancy across the genomic sequences for the irradiated than unirradiated DNAs (Figure 3), indicating significantly reduced DNA-binding specificity. The observed reduction in sequence specificity in the context of UVC irradiation suggests that TFAM occupancy on UVC-irradiated DNA outcompetes any sequence specificity it has, at least in the absence of other proteins. These data would represent an extensively damaged genome because the DNA was irradiated at a high dose of UVC, 1500 J/m<sup>2</sup>. In contrast, the DNA in the AFM experiments contain on average approximately three lesions per plasmid. Strikingly, the significant increase in compaction as evidenced by the increase in punctate structures of pUC19 by TFAM in the presence of UVC irradiation (Figure 4D) suggests that only a few lesions are sufficient to disrupt the DNA-binding specificity of TFAM and drive dramatic compaction of pUC19.

In our AFM images, we observe tracts of protein binding along the DNA that bridge two regions of DNA together (Figure 4A, 4C, S17, S18) in a zipper-like manner, without inducing any significant supercoiling. These results are consistent with previous studies (9, 10). In contrast, the X-ray crystal structures of TFAM bound to DNA show the protein wrapped around the DNA, forming a U-turn bend in the DNA (30, 33, 34). This mode of binding to plasmid DNA would induce supercoiling (68). Taken together, these results indicate that TFAM can bind DNA in at least two different states, one which induces a U-turn bend in the DNA and one that does not dramatically distort the DNA. In the AFM images, in general, we observe a single cluster of TFAM per plasmid, with or without adjacent fiber formation. In the absence of UV irradiation, the clusters occupy a smaller region of DNA and exist in looser, more intermediate states (Figure 4D), whereas in the irradiated DNA, TFAM forms more punctate clusters that contain more of the plasmid DNA (Figure 4D). One possible explanation for how only a few lesions could promote increased compaction of the DNA by TFAM is that TFAM may preferentially bind UV damage, which distorts DNA, in the U-turn mode. This U-turn in the DNA would bring two DNA double strands close to one another and therefore promote local zipping up of the DNA. Subsequently, these clusters may cooperatively assemble together to form tight compact structures (Figure 4C, 4E).



## Conclusions and future directions

Overall, we found that cellular responses to mtDNA damage including increased mtDNA degradation, upregulation of mtDNA replication machinery, upregulated TFAM, and increased mtRNA transcription occur in the absence of loss of mitochondrial membrane potential or changes to ATP levels. We identified TFAM as a potential DNA damage sensing protein in that it promotes UVC-dependent conformational changes in the nucleoids *in vitro*, making them more compact. The increased compaction may serve to signal damage in the mitochondrial genome. For example, once “tagged” by TFAM, damaged mtDNAs might be actively removed via mitophagy (23, 27), autophagy (22), extracellular export, TFAM-mediated nucleoid-phagy (69), or by more recently-identified non-autophagic pathways (70, 71). Indeed, we found that TFAM knockdown alters the levels of mtDNA damage over time. Alternately, simply by removing these damaged genomes from the pool that is actively used for transcription and replication by making them inaccessible to other proteins, TFAM-mediated compaction could serve to store damaged genomes until they are removed passively by normal mtDNA turnover processes. In any case, these non-repair pathways for removing mtDNA damage may provide a mechanism for preventing mtDNA mutagenesis. These possibilities are illustrated schematically in Figure 6, which depicts a hypothetical model that will require additional work to test *in vivo*. Overall, this work presents a first step towards understanding how TFAM may play a role in sensing and responding to irreparable mtDNA damage.

## Experimental Procedures

### Cell Culture

HeLa cells were kindly provided by Dr. Chantell Evans (Duke University). TFAM-HA TetOn HeLa S3 cells were kindly provided by Dr. Stefan Isaac and Dr. Stirling Churchman (Harvard University). Cells were grown in DMEM containing glucose and pyruvate (Thermo Fisher Scientific 11995073) supplemented with 10% Fetal Bovine Serum (Thermo Fisher Scientific 10437028), 1% penicillin/streptomycin and maintained at 37°C with 5% CO<sub>2</sub> in a humidified chamber. Cells were routinely tested for mycoplasma contamination and authenticated using STR profiling by the Duke University Cell Culture Facility.

### UVC Exposures in Cell Culture

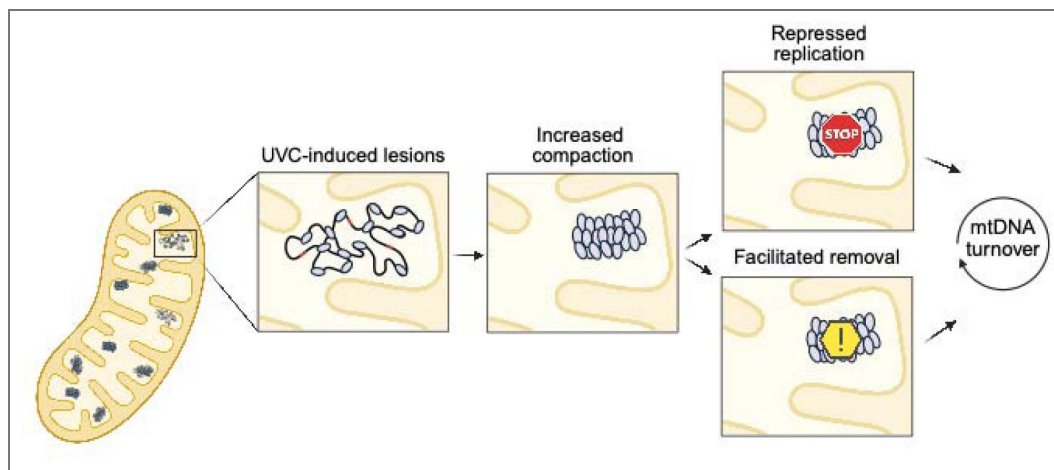
HeLa cells were grown to 50% confluency. At the time of exposure, media was removed, cells were washed once in PBS, aspirated to remove as much of the PBS as possible to facilitate the effects of the UVC irradiation, and then exposed to UVC using an ultraviolet lamp with built-in UVC sensor (CL-1000 Ultraviolet Crosslinker, UVP, Upland, CA, USA) with peak emission at 254 nm. Following exposure (~1 minute), cell culture media was immediately replaced.

### Cell Viability

To assess cell viability, 10,000 cells were plated in black clear-bottom 96-well plates and allowed to adhere overnight. Cells were washed with PBS and then dosed with UVC as described in the previous section. Experiments were performed in triplicate with three biological replicates completed for all doses. The 0-, 24-, and 48-hour dose responses were analyzed at their respective time points. Cell viability was assessed using the alamarBlue HS Cell Viability Reagent (Invitrogen) and a FLUOstar Optima microplate reader. The data was analyzed via two-way ANOVA with a Dunnett’s correction for multiple comparisons to the control for each time point.

### Gene Expression

Quantitative real-time RT-PCR was performed for *TFAM*, *POLG*, *NRF-1*, *PGC1LJ*, *POLRMT*, and *ND-1*. RNA was extracted from HeLa cells using an RNeasy Kit (QIAGEN) and stored at -80°C until analysis. qPCR was performed using the qTower<sup>3</sup> Real-Time PCR System (Analytik-Jena). The *GAPDH* Endogenous Control (Taqman probe ID: Hs02786624\_g1, VIC-MGB) and either *TFAM*



**Figure 6.** Hypothesized cellular model of the role of TFAM in mtDNA damage sensing.

The *in vitro* data provided in this study indicate that TFAM more readily compacts DNA harboring UV-induced lesions. While there is currently no *in vivo* evidence to suggest damaged mtDNA is more compacted, future work should determine whether this phenomenon is occurring in cells. *In vivo*, this feature may serve to “tag” mitochondrial genomes as damaged, which could lend itself to repression of the replication of damaged genomes, flagging damaged genomes for targeted degradation, or both. Both active removal and repression of replication would allow for removal of damaged genomes during mtDNA turnover processes and provide a mechanism for preventing mtDNA mutagenesis. Created with [BioRender.com](https://www.biorender.com).

(Taqman probe ID: Hs00273372\_s1, FAM-MGB), *POLG* (Taqman probe ID: Hs00160298\_m1, FAM-MGB), *NRF-1* (Taqman probe ID: Hs00602161\_m1, FAM-MGB), *PGC1LJ* (Taqman probe ID: Hs00173304\_m1, FAM-MGB), *POLRMT* (Taqman probe ID: Hs04187596\_g1, FAM-MGB), or *ND-1* (Taqman probe ID: Hs02596873\_s1, FAM-MGB) probes were multiplexed in a 1:1 ratio for each sample. Results for *GAPDH* were used to normalize for RNA input. PCR reactions each contained 100 ng RNA, 0.5  $\mu$ L of each probe, 5  $\mu$ L of Quantabio qScript one-step RT-qPCR ToughMix (QuantaBio) and was brought to a total reaction volume of 10  $\mu$ L with nuclease-free water. Samples were run in triplicate and PCR cycling conditions were as follows: 50°C for 10 minutes, followed by 95°C for one minute, followed by 40 cycles of 95°C for 10 seconds and 60°C for one minute. To examine expression of each gene, the delta delta CT method was used to calculate fold change in gene expression. The data were analyzed via two-way ANOVA with Dunnett's post-hoc test for multiple comparisons.

## Mitochondrial Membrane Potential

HeLa cells were plated and allowed to attach and proliferate for 24 hours. Then, cells were exposed to either 0, 10, 30, or 50 J/m<sup>2</sup> of UVC. Cells were harvested and stained to acquire by flow cytometry at 6 and 24 hours post UVC exposure. Each UVC dose had a tetramethylrhodamine, methyl ester (TMRM)-stained sample and an unstained control. Non-UVC-exposed cells from each timepoint had additional TMRM-stained tubes for carbonyl cyanide-p-trifluoromethoxyphenylhydrazone (FCCP) positive controls. Cells were aliquoted (4.0 X 10<sup>5</sup> per sample) and washed with PBS. Samples were stained in 1 mL of Live Cell Imaging Medium with a final concentration of 15 nM TMRM for 20 minutes. TMRM was left in cell suspension during flow acquisition. Samples were acquired on a BD FACSCanto-II flow cytometer using the blue laser (488 nm) and the 585/42 detection filter to excite and detect TMRM. Events were gated to include live, single cells, and 10,000 events were acquired from this population.

Immediately before their acquisition, TMRM-stained control cells were spiked with 5  $\mu$ M FCCP. The FCCP positive control was incubated for 5 min before acquiring. In FlowJo, live cells were manually gated in a FSC-A vs SSC-A plot, then single cells were gated in a FSC-A vs FSC-H plot. Median fluorescence intensity (MFI) was tabulated for the histogram of each sample. Background fluorescence from the unstained sample of each UVC dose was subtracted from the corresponding stained samples. The average and standard error of three experiments were plotted. Data was analyzed via two-way ANOVA for treatment and timepoint, with a Dunnett's correction for multiple comparisons to the control for each time point.

## ATP Quantification

To determine ATP concentrations, 10,000 HeLa cells were plated in white, flat-bottom 96-well plates. Cells were exposed to UVC irradiation as described above. At 6 or 24 hours post-UVC exposures, the CellTiter-Glo assay (Promega) was performed according to the manufacturer's instructions. Briefly, plates were equilibrated to room temperature before addition of 100  $\mu$ L of reconstituted assay reagent.

Cells were lysed via orbital shaking (200 rpm for 2 minutes) followed by a 10-minute incubation at room temperature. Luminescence was measured on a FLUOstar Optima microplate reader. A standard curve was prepared fresh before each assay with ATP disodium salt (Sigma) in complete DMEM. The data was analyzed via two-way ANOVA with a Dunnett's correction for multiple comparisons to the control for each time point.

## Live Cell Imaging

HeLa cells (1.5 X 10<sup>3</sup>) were seeded onto 35 mm glass bottom well petri dishes (Mattek) and allowed to adhere. After 48 hours in culture, cells were washed once with PBS and then exposed to either 0 or 10 J/m<sup>2</sup> UVC as described previously. Cells were then immediately incubated with picogreen at a concentration of 1  $\mu$ L/mL of medium for 1.5 hours to stain mtDNA. Cells were then washed three times and given fresh media. Twenty-four hours after the UVC exposure, cells were incubated with lysotracker red (75 nM), mitotracker deep red (150 nM), and 7  $\mu$ g/mL Hoescht 33342 for 30

minutes, again followed by three quick washes, and then placed in imaging medium. Cells were imaged in an environmental chamber at 37°C in an Andor Dragonfly 505 unit with Borealis illumination spinning disk confocal 100x/1.40-0.70 HCX PL APO (Leica 11506210) oil-immersion objective and an Andor iXon Life 888 1024x1024 EMCCD camera.

## Confocal Image Analysis

Images were deconvoluted using Huygens SVI deconvolution software. For each channel, images were segmented in Labkit (72). Segmented images were used to quantify area of the cell, area of mitochondria, mitochondrial morphology parameters, number of picogreen spots, and number of lysosome spots. To assess lysosome colocalization, object-oriented colocalization (73) was performed on segmented images using a custom Python script. At least 30 cells in each treatment group were analyzed from three independent biologic replicates. The data were analyzed via two-tailed unpaired t-test.

## Quantification of DNA Damage

To determine the lesion frequency associated with mitochondrial DNA, a quantitative long-range PCR assay was used (74). For cell samples, DNA was isolated from frozen cell pellets using genomic tips (Qiagen). QPCR was run on the samples to generate a long product (~10kb) as well as a short product (~200 bp). Given the ability for DNA damage to block the DNA polymerase used in this PCR, the amount of damage is then calculated based on the amount of amplification. The long product is used to calculate reduced amplification compared to the controls, whereas the short product is used to normalize sample to sample variability in mitochondrial DNA copy number. The data were analyzed via two-way ANOVA with a Dunnett's post-hoc test for multiple comparisons.

## TFAM Overexpression

TFAM-HA TetOn HeLa S3 cells were grown in six-well dishes to a confluency of 50% and treated with 100 ng/mL doxycycline for 48 hours. Cells were provided fresh media every 24 hours for the duration of the experiment. UVC exposure was performed as described above with cells exposed to either 0, 10, 30, 50, 70, or 100 J/m<sup>2</sup>. mtDNA damage was quantified at 0 hours, 24 hours, and 48 hours following UVC exposure as described previously and cell viability was assessed. TFAM overexpression levels in control samples were confirmed by real-time PCR and western blot analysis. Real-time PCR to assess *TFAM* gene expression was performed as described above.

For western blot analysis, cells were washed twice in ice-cold PBS, and then lysed using RIPA buffer containing a protease inhibitor cocktail. Protein content was determined using a BCA Assay (Pierce). Protein lysate (30 µg/well) was loaded into a 10% SDS-PAGE gel. Gels were transferred onto activated PVDF membrane, followed by immunoblot. Densitometry was performed using ImageJ. TFAM was normalized to β-actin and samples were compared between doxycycline treatment groups at their respective time points. Antibodies used were anti-TFAM antibody (Santa Cruz sc-376672) and an anti-β-Actin antibody (Sigma A5441). The data were analyzed via two-way ANOVA.

## TFAM Knockdown

HeLa cells were reverse transfected in 6-well plates with 8.3 pmol siRNA (TFAM Flexitube siRNA, GeneGlobe ID SI04988487; or AllStars Negative Control siRNA, Qiagen) and 2.5 µL Lipofectamine RNAiMAX (Invitrogen) per well according to the manufacturer's instructions. Cell plating densities were optimized to avoid confluency for each timepoint and scaled up to harvest sufficient cells for analysis. At 24 hours post-transfection, cells were exposed to either 0 J/m<sup>2</sup> or 30 J/m<sup>2</sup> UVC and harvested immediately (timepoint 0 hr) or at 24 or 48 hours post-UVC. Harvested cells were washed with PBS, pelleted, and frozen at -80°C prior to DNA extraction. Data was analyzed via three-way ANOVA with a Dunnett's post-hoc test for multiple comparisons.

## Purification of Human TFAM

Human\_TFAM\_NoMTS\_pET28 was a gift from David Chan (Addgene plasmid # 34705; <http://n2t.net/addgene:34705> ; [RRID:Addgene\\_34705](https://doi.org/10.12555/RRID:Addgene_34705)). Recombinant human TFAM was purified as previously described (33), flash frozen in liquid nitrogen and stored at -80°C. Protein concentration was determined using a Qubit Protein BR Assay (Thermo A50668).

## Generation of Substrates for Atomic Force Microscopy

To assess alterations in TFAM compaction of DNA in the context of UVC-induced lesions, we assessed TFAM binding to circular pUC19. Levels of UVC exposure were chosen based on previous dose response curves performed on naked mtDNA indicating a linear increase in lesions associated with increasing UVC exposure, where 10 J/m<sup>2</sup> results in roughly 1 lesion per 10kb (75). Given the similar content of adjacent pyrimidine bases (whole mitochondrial genome: 535/kb; 8.9kb product used in mtDNA damage assay: 532/kb; and pUC19: 510/kb), we exposed the pUC19 (2.686 kb in size) to 100 J/m<sup>2</sup> to induce roughly three lesions per DNA molecule.

## Atomic Force Microscopy

Topoisomerase I (NEB # M0301S) was used to relax all pUC19 DNA substrates used for AFM TFAM protein-DNA experiments. Each reaction was carried out by using 1 µg of DNA (either UV-irradiated or undamaged), 10x CutSmart Buffer, and 1 unit of Topoisomerase I diluted to a total reaction volume of 50 µL. The reaction was then placed in a thermocycler and ran at 37°C for 15 minutes, and afterwards a PCR cleanup kit (QIAGEN) was used to remove protein and other impurities before performing AFM experiments.

Undamaged or damaged DNA (1.5 ng/µL) was incubated for two minutes on ice with varying concentrations of TFAM in 25 mM HEPES, 10 mM Mg(OAc)<sub>2</sub>, 25 mM NaOAc, 75 mM K(OAc), pH 7.5, then deposited onto ethanolamine treated mica and imaged in air using a JPK NanoWizard 4 XP in tapping mode. Mica was treated with ethanolamine by vapor deposition in a desiccator by aliquoting 20 µL of ethanolamine stock onto a piece of parafilm and placing it into the desiccator with freshly peeled mica for 15 minutes to allow time for the ethanolamine to vaporize and transfer onto the mica surface. Cantilevers used were PPP-NCHR (Nanoworld, Matterhorn, Switzerland) with nominal resonance frequencies of 330 kHz. Images were taken at a 2.0 x 2.0 µm area for small DNA fragments such as pUC19 while the full-length mitochondrial DNA images were between 2-5 µm. Images of TFAM and small DNA oligonucleotides were taken at a 1.0 x 1.0 µm area on untreated mica. All AFM images were taken with a resolution of 512 x 512 pixels, and a scan rate of 2 Hz. Additionally, each TFAM-DNA complex with pUC19 was categorized into three different categories: dispersed, intermediate, or punctate, based on the degree of observable DNA compaction and height of the TFAM nucleoid structure.

Images were analyzed using the MATLAB program ImageMetrics (<https://imagemetrics.wordpress.com/>) in MATLAB version R2020b (MathWorks, Natick MA). Images were line-wise flattened, and subsequently, a threshold was applied to isolate the individual protein only or protein-DNA complexes on the surface. For each of these complexes or proteins, the volume, area, and height were recorded. Histograms of the volumes of TFAM protein alone or in the presence of DNA were generated from the data using MATLAB version R2020b.

We used AFM to determine the oligomerization state of TFAM bound to some DNA oligonucleotides (33 base pairs) used for the fluorescence anisotropy studies. TFAM (30 nM) was incubated on ice in the absence or presence of DNA (10 nM) for two minutes, deposited on the surface, rinsed, and dried with N<sub>2</sub>. ImageMetrics was used to determine the volumes of the individual particles on the surface as described earlier. The AFM volume of proteins are directly proportional the protein molecular weight allowing determination of the oligomerization state (58–60, 76). The oligomerization states are determined from histogram plots of the distribution of volumes that were generated in MATLAB version R2020b.



## In vitro Nucleoid Reconstitution

Full-length linear mtDNA was generated by long-range PCR amplification using the following primers; F, 5′ – GGT TCA GCT GTC TCT TAC TTT TAA CCA GTG – 3′ and R, 5′ – CTC GTG GAG CCA TTC ATA CAG GTC – 3′. Genomic DNA was isolated from HeLa cells using the Qiagen QiAmp DNA Mini Kit according to the manufacturer's instructions and used as template DNA. PCR reactions were carried out using 100 ng template in 50  $\mu$ L reactions with LongAmp Hot Start Taq Polymerase (New England Biosciences). DNA was amplified using a two-step amplification protocol: 95°C for 2 minutes, 30 cycles of 95°C for 20 seconds and 68°C for 13 minutes, and a final extension at 68°C for 18 minutes. PCR product was then purified using a Zymo Genomic DNA Clean & Concentrator kit (Zymo Research D4011) and the DNA concentration was determined using a Nanodrop One (Thermo Scientific).

*In vitro* nucleoids were generated in 50  $\mu$ L reactions, containing 20 ng/ $\mu$ L full-length linear mtDNA with 0, 100, 250, or 1000 nM TFAM (added last) in binding buffer (25 mM HEPES pH 7.5, 75 mM KOAc, 25 mM NaOAc, 10 mM Mg(OAc)<sub>2</sub>). The degree of compaction of the mtDNA at these TFAM concentrations was confirmed using AFM as described earlier. After the addition of TFAM, reactions were incubated on ice for two minutes, then deposited as a 50  $\mu$ L drop onto a petri dish lid and then immediately exposed to either 0, 5, 10, 30, or 50 J/m<sup>2</sup> UVC. Following UVC treatment, a Zymo Genomic DNA Clean & Concentrator 10 kit (Zymo Research D4011) was used to remove the protein, and the DNA concentration was determined using a Nanodrop One (Thermo Scientific). Samples were diluted to 25 pg/ $\mu$ L and levels of mtDNA damage were quantified as described above.

## High-throughput, High-resolution Binding of TFAM to the Mitochondrial Genome

High-density DNA chips (or arrays) printed with oligos covering the entire mitochondrial genome at a 2-bp resolution were used to measure TFAM binding in the presence and absence of UV-induced damage. Single-stranded DNA arrays were purchased from Agilent in a 4 × 180 k format containing four identical chambers with 176,024 DNA spots in each chamber. The DNA library synthesized on the array contained the entire mitochondrial genome in 33-nt variable sequences and a 27-nt primer binding sequence. A sliding window approach was used to partition the hg38 human mitochondrial genome into overlapping 33-nt regions, where each sequential window was shifted by 2 nt along the mtDNA sequence (the single 'N' was replaced with a cytosine). For every mtDNA window, an additional DNA sequence was generated using the reverse complement of the window, such that the full genome was synthesized *de novo* in both the 5′ to 3′ and 3′ to 5′ orientations on the glass slide. TFAM binding to a universal DNA library containing all possible 7-mers (53) was performed prior to library designs to determine the range of binding specificity, optimize the final protein concentration to be used, and select low affinity non-mitochondrial probes to include in the sequence library for normalization purposes. The number of replicate spots for each sequence on the array was 10. In total, there were 8,285 unique probes in each orientation on the array (16,570 total), corresponding to the mitochondrial genome.

The DNA was double stranded on the arrays via primer extension, as described previously (53), and then subjected to 1,500 J/m<sup>2</sup> UVC irradiation to damage DNA in two of the chambers on the array, as previously described (77). Finally, purified TFAM as well as Penta-His Alexa 488 (Qiagen) were incubated in the chambers in TFAM binding buffer (25 mM HEPES, 10 mM Mg(OAc)<sub>2</sub>, 100 mM NaOAc, pH 7.5) as previously described (78) and the fluorescent signal associated with bound protein for each DNA spot was determined using a GenePix® 4400A microarray scanner and the GenePix® Pro 7.3 software. The assay was performed using a final concentration of either 30 nM or 300 nM TFAM. In total, assays were run at 30 nM TFAM with and without UVC as well as 300 nM TFAM with and without UVC.

Fluorescence intensity values for each spot were normalized via detrending, as previously described (78). Mean, median, standard deviation, and standard error were calculated from normalized fluorescence intensity values across the 10 replicates for each spot. Kernel density

estimates were calculated for the fluorescence values of the bottom universal binders for the conditions in each chamber, and a Gaussian curve was fit to those data. For each chamber, the mean and standard deviation estimates from the Gaussian fit were used to convert all median fluorescence intensity values into TFAM binding z-scores, which reflect the DNA binding specificity of TFAM under that experimental condition.

Given that the assay was performed on sequences generated in 5' to 3' orientation as well as a 3' to 5' orientation, the maximum z-score obtained from each orientation (*i.e.*, whichever was higher) was used to generate binding plots. Plots were then smoothed using a 22-nt window, in order to account for the 22-nt footprint of TFAM binding. Most binding z-scores are expected to be low, regardless of the binding conditions, as most sequences are expected to show some low background level of binding. However, under conditions where TFAM binds DNA with high specificity, we expect the array-based assay to also identify sequences with high z-score, *i.e.*, sequences that TFAM can bind with higher occupancy, and distinguish them from background. In contrast, conditions where all z-scores are generally low are indicative of low binding specificity throughout, as all sequences are bound similarly by TFAM, and no particular target sites stand out. All statistics were performed using a custom Python notebook.

## Fluorescence Polarization

Sequences representing a broad range of z-scores from the array-based experiments (Table S2) were ordered from Integrated DNA Technologies as HPLC-purified, double-stranded oligos (reverse complement not shown in table) with a 5' 6-carboxyfluorescein modification on the strand listed. Each sequence was tested with and without 1,500 J/m<sup>2</sup> UVC irradiation to match conditions used in the array-based assay. Each condition was tested in triplicate.

TFAM was serially diluted in binding buffer (25 mM HEPES, 10 mM Mg(OAc)<sub>2</sub>, 25 mM NaOAc, 75 mM K(OAc), pH 7.5) from 15 to 0 nM on ice. Dilutions were then mixed in a low-binding Costar 96-well half-area black well plate (Corning, Corning, NY) with DNA diluted in binding buffer, for a final DNA concentration of 1 nM and volume of 55 µL in each well. Plates were read with a PHERAstar microplate reader from BMG Labtech (Cary, NC) using an FP filter with excitation at 485 nm and emission at 590 nm. Anisotropy was calculated with Equation 1:

$$r = \frac{I_{\parallel} - I_{\perp}}{I_{\parallel} + 2I_{\perp}} \quad \text{Eq. 1}$$

where  $r$  is anisotropy and  $I_{\parallel}$  and  $I_{\perp}$  are fluorescence intensity in the parallel and perpendicular direction, respectively. Anisotropy was then plotted against TFAM concentration and fit to Equation 2:

$$r = r_f + \frac{r_b \cdot [\text{TFAM}]^n}{K_D^n + [\text{TFAM}]^n} \quad \text{Eq. 2}$$

where  $r_f$  is the fraction of signal from free DNA,  $r_b$  is the fraction of signal from bound DNA,  $[\text{TFAM}]$  is TFAM concentration,  $n$  is the Hill coefficient, and  $K_D$  is the dissociation constant. The variance  $\sigma^2$  associated with each of the derived  $K_D$  and  $n$  values from individual fits was identified from the diagonal of the covariance matrix, and the inverse-variance weighted mean  $\bar{x}_{wtd}$  and standard error  $(\sigma_{\bar{x}})_{wtd}$  were calculated using Equations 3 and 4, respectively, across the three experimental replicates:

$$\bar{x}_{wtd} = \frac{\sum_{i=1}^m x_i / \sigma_i^2}{\sum_{i=1}^m 1 / \sigma_i^2} \quad \text{Eq. 3}$$

$$(\sigma_{\bar{x}})_{wtd} = \sqrt{\frac{\left( \frac{\sum_{i=1}^m x_i^2 / \sigma_i^2}{\sum_{i=1}^m 1 / \sigma_i^2} - \bar{x}_{wtd}^2 \right) \left( \frac{m}{m-1} \right)}{m}} \quad \text{Eq. 4}$$

where  $x_i$  is the derived value from each fit,  $m$  is the number of replicates in the condition (79). Curve fitting and statistical analysis were performed in Python.

## Data Availability

All the raw and processed protein-DNA binding assay data generated and used in this study are located on NCBI Gene Expression Omnibus (GEO) [GSE281949]. Custom python notebooks are available on GitHub (<https://github.com/satusky/>) in the tfam-mtdna-uv-array repository. All other data are available in the Duke Research Data Repository (<https://doi.org/10.7924/r40s0211r>).

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## Additional information

### Author Contributions

DEK: conceptualization, investigation, formal analysis, writing – original draft, EEB: investigation, formal analysis, writing – original draft, MJS: investigation, formal analysis, visualization, writing – original draft, AG: investigation, formal analysis, IR: investigation, formal analysis, CJ: investigation, formal analysis, ELD: investigation, formal analysis, YZ: investigation, formal analysis, WZ: investigation, formal analysis, HW: investigation, formal analysis, EC: investigation, formal analysis, SKM: funding acquisition, supervision, DE: conceptualization, funding acquisition, supervision, writing – original draft, RG: conceptualization, funding acquisition, supervision, writing – original draft, JNM: conceptualization, funding acquisition, supervision, writing – original draft.

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### Abbreviations

AFM: atomic force microscopy  
mtDNA: mitochondrial  
DNA NER: nucleotide excision repair  
TFAM: Transcription Factor A, Mitochondrial  
UVC: ultraviolet-C irradiation

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## Additional files

[Supplemental Figures](#) 

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## Peer reviews

### Reviewer #1 (Public review):

#### Summary:

The authors investigate how UVC induced DNA damage alters the interaction between the mitochondrial transcription factor TFAM and mtDNA. Using live-cell imaging, qPCR, atomic force microscopy (AFM), fluorescence anisotropy, and high-throughput DNA-chip assays, they show that UVC irradiation reduces TFAM sequence specificity and increases mtDNA compaction without protecting mtDNA from lesion formation. From these findings the authors suggest that TFAM acts as a "sensor" of damage rather than a protective or repair-promoting factor.

#### Strengths:

- (1) The focus on UVC damage offers a clean system to study mtDNA damage sensing independently of more commonly studied repair pathways, such as oxidative DNA damage. The impact of UVC damage is not well understood in the mitochondria and this study fills that gap in knowledge.
- (2) In particular, the custom mitochondrial genome DNA chip provides high resolution mapping of TFAM binding and reveals a global loss of sequence specificity following UVC exposure.
- (3) The combination of in vitro TFAM DNA biophysical approaches combined with cellular responses (gene expression, mtDNA turnover) provides a coherent multi-scale view.
- (4) The authors demonstrate that TFAM induced compaction does not protect mtDNA from UVC lesions, an important contribution given assumptions about TFAM providing protection.

#### Weaknesses:

- (1) The authors show a decrease in mtDNA levels and increased lysosomal colocalization but do not define the pathway responsible for degradation. Distinguishing between replication dilution, mitophagy, or targeted degradation would strengthen the interpretation and justifies future experiments.
- (2) The manuscript briefly notes enrichment of TFAM at certain regions of the mitochondrial genome but provides little interpretation of why these regions are favored. Discussion of whether high-occupancy sites correspond to regulatory or structural elements would add valuable context.
- (3) The authors provide a discrepancy between the anisotropy and binding array results. The reason for this is not clear and one wonders if an orthogonal approach for the binding experiments would elucidate this difference (minor point).

#### Assessment of conclusions:

The manuscript successfully meets its primary goal of testing whether TFAM protects mtDNA from UVC damage and the impact this has on the mtDNA. While their data points to an intriguing model that TFAM acts as a sensor of damaged mtDNA, the validation of this model requires further investigation to make the model more convincing. This is likely warranted

for a followup study. Also the biological impact of this compaction, such as altering transcription levels is not clear in this study.

Impact and utility of the methods:

This work advances our understanding of how mitochondria manage UVC genome damage and proposes a structural mechanism for damage "sensing" independent of canonical repair. The methodology, including the custom TFAM DNA chip, will be broadly useful to the scientific community.

Context: The study supports a model in which mitochondrial genome integrity is maintained not only by repair factors, but also by selective sequestration or removal of damaged genomes. The demonstration that TFAM compaction correlates with damage rather than protection reframes an interesting role in mtDNA quality control.

Comments on revised version:

The authors addressed all concerns during the revision.

<https://doi.org/10.7554/eLife.108862.2.sa3>

## Reviewer #2 (Public review):

Summary:

King et al. present several sets of experiments aimed to address potential impact of UV irradiation on human mitochondrial DNA as well as possible role of mitochondrial TFAM protein in handling UV irradiated mitochondrial genomes. The carefully worded conclusion derived from the results of experiments performed with human HeLa cells, in vitro small plasmid DNA, with PCR-generated human mitochondrial DNA and with UV-irradiated small oligonucleotides is presented in the title of the manuscript: "UV irradiation alters TFAM binding to mitochondrial DNA". Authors also interpret results of somewhat unconnected experimental approaches to speculate that "TFAM as a potential DNA damage sensing protein in that it promotes UVC-dependent conformational changes in the [mitochondrial] nucleoids, making them more compact. They further propose that such a proposed compaction might trigger removal of UV-damaged mitochondrial genomes as well as facilitates replication of undamaged mitochondrial genomes.

Strengths:

(1) Authors presented convincing evidence that a very high dose (1500 J/m<sup>2</sup>) of UVC applied to oligonucleotides covering the entire mitochondrial DNA genome alleviates sequence specificity of TFAM binding (Figure 3). This high dose was sufficient to cause UV-lesions in a large fraction of individual oligonucleotides. The method has been developed in the lab of one of the corresponding authors (ref. 74) and is technically well refined. This result can be published as is or in combination with other data.

(2) Manuscript also presents AFM evidence (Figure 4) that TFAM, which was long known to facilitate compaction of mitochondrial genome (Alam et al., 2003; PMID 12626705 and follow up citations), causes in vitro compaction of a small pUC19 plasmid and that approximately 3 UVC lesions per plasmid molecule results in slight albeit detectable increase in TFAM compaction of the plasmid.

Both results are discussed in line of a possible extrapolation to in vivo phenomena. The revised version of the discussion includes a clear statement that no in vivo support was provided within the set of experiments presented in the manuscript.

Weaknesses:



The experiments presented on Figures 3 and 4 may support the speculation that TFAM can carry protective role of eliminating mitochondrial genomes with bulky lesions by way of excessive compaction and removal damaged genomes from the in vivo pool, however extensive additional studies that would go well beyond the experiments described in this paper are needed to fill the gap between this set of results and the proposed explanations.

<https://doi.org/10.7554/eLife.108862.2.sa2>

### Reviewer #3 (Public review):

#### Summary

The study is grounded in the observation that mitochondrial DNA (mtDNA) shows some resistance to mutagenesis under genotoxic stress. The manuscript focuses on the effects of UVC-induced DNA damage on TFAM-DNA binding in vitro and in cells. The authors demonstrate increased TFAM-DNA compaction following UVC irradiation in vitro, as assessed by high-throughput protein-DNA binding assays and atomic force microscopy (AFM). The authors did not observe a similar trend in fluorescence polarization assays and attributed the difference in the extent of TFAM oligomerization as a potential reason. In cells, the authors found that UVC exposure increased mRNA levels of TFAM, POLG, and POLRMT without altering mitochondrial membrane potential. Overexpressing TFAM in cells or varying TFAM concentration in reconstituted nucleoids did not alter the accumulation or disappearance of mtDNA damage. Based on their data, the authors proposed a plausible model: following UVC-induced DNA damage, TFAM facilitates nucleoid compaction, which may signal damage in the mitochondrial genome. The proposed model may inspire future follow-up studies to further study the role of TFAM in sensing UVC-induced damage.

#### Comments on revised version:

The authors have addressed the reviewer's concerns.

<https://doi.org/10.7554/eLife.108862.2.sa1>

### Author response:

The following is the authors' response to the original reviews.

#### **Public Reviews:**

##### **Reviewer #1 (Public review):**

#### *Summary:*

*The authors investigate how UVC-induced DNA damage alters the interaction between the mitochondrial transcription factor TFAM and mtDNA. Using live-cell imaging, qPCR, atomic force microscopy (AFM), fluorescence anisotropy, and high-throughput DNA-chip assays, they show that UVC irradiation reduces TFAM sequence specificity and increases mtDNA compaction without protecting mtDNA from lesion formation. From these findings, the authors suggest that TFAM acts as a "sensor" of damage rather than a protective or repair-promoting factor.*

#### *Strengths:*

*(1) The focus on UVC damage offers a clean system to study mtDNA damage sensing independently of more commonly studied repair pathways, such as oxidative DNA*

damage. The impact of UVC damage is not well understood in the mitochondria, and this study fills that gap in knowledge.

(2) In particular, the custom mitochondrial genome DNA chip provides high-resolution mapping of TFAM binding and reveals a global loss of sequence specificity following UVC exposure.

(3) The combination of in vitro TFAM DNA biophysical approaches, combined with cellular responses (gene expression, mtDNA turnover), provides a coherent multi-scale view.

(4) The authors demonstrate that TFAM-induced compaction does not protect mtDNA from UVC lesions, an important contribution given assumptions about TFAM providing protection.

#### Weaknesses:

(1) The authors show a decrease in mtDNA levels and increased lysosomal colocalization but do not define the pathway responsible for degradation. Distinguishing between replication dilution, mitophagy, or targeted degradation would strengthen the interpretation

We thank the reviewer for their careful reading of our manuscript and thoughtful suggestions. We agree that distinguishing between replication dilution, mitophagy, and/or targeted degradation would strengthen our understanding of how UV-induced DNA damage is handled in the mitochondria. Currently we are undertaking experiments to tease this apart, but consider the scope of those experiments to be beyond this manuscript and expect to publish them in a subsequent paper rather than this one. We added text explicitly stating that these possibilities are not distinguished by our results in pages 8-9 in the Discussion under the subsection ‘Mitochondria respond to UVC-induced mtDNA damage in the absence of apparent mitochondrial dysfunction’.

(2) The sudden induction of mtDNA replication genes and transcription at 24 h suggests that intermediate timepoints (e.g., 12 hours) could clarify the kinetics of the response and avoid the impression that the sampling coincidentally captured the peak.

We agree and have added additional timepoints of 12 hours and 18 hours post exposure. We have updated Figure 2 to include the new data and have added text on page 4 to include these results.

(3) The authors report no loss of mitochondrial membrane potential, but this single measure is limited. Complementary assays such as Seahorse analysis, ATP quantification, or reactive oxygen species measurement could more fully assess functional integrity.

We focused on membrane potential because loss of membrane potential is such a well-understood of mechanism for triggering mitophagy, but agree that these additional measurements are useful. We have added experiments to assess ATP levels, but did not see changes; we have added this data to Figure 2. We have also added text highlighting that we previously assessed mtROS following the same levels of UV exposure and observed no changes (in the results section on page 5 and in the discussion section on page 9). Given that we observe no changes in membrane potential or ATP, we have opted to not move forward with Seahorse analysis for the purposes of this paper.

(4) The manuscript briefly notes enrichment of TFAM at certain regions of the mitochondrial genome but provides little interpretation of why these regions are favored. Discussion of whether high-occupancy sites correspond to regulatory or structural elements would add valuable context.

We agree a discussion of these findings provides context and insight into where the field is currently in understanding TFAM sequence specificity. We have updated text in the discussion (pages 9-10) to include our thoughts on the drivers of TFAM sequence specificity with regard to the discrepancy with the anisotropy data and the lack of overlap with regulatory/structural elements.

*(5) It remains unclear whether the altered DNA topology promotes TFAM compaction or vice versa. Addressing this directionality, perhaps by including UVC-only controls for plasmid conformation, would help disentangle these effects if UVC is causing compaction alone.*

We have added an additional control making this comparison and updated the text on page 7 in the results section. UVC by itself (without TFAM being present) does not alter the plasmid compaction; see new supplemental Figure S16.

*(6) The authors provide a discrepancy between the anisotropy and binding array results. The reason for this is not clear, and one wonders if an orthogonal approach for the binding experiments would elucidate this difference (minor point).*

The discrepancy between anisotropy and the binding array results is certainly unusual and contrary to previous studies that have used these arrays. In addition to the anisotropy experiments, we selected a 'high occupancy' and 'low occupancy' sequence from the binding array and performed oligomerization experiments using atomic force microscopy, which allowed us to detect small changes in cooperativity (see supplemental Figure S15). We previously only discussed this briefly in the results section on page 6, but we have now updated the discussion section (pages 9-10) to highlight this finding and put forth ideas for the field as to why we think this might be the case. While we do see that the binding array data aligns with oligomerization and cooperativity of TFAM, we still do not know what it is about these sequences that would drive such differences in TFAM binding, but we speculate that it could have something to do with flexibility of the DNA sequences.

#### *Assessment of conclusions:*

*The manuscript successfully meets its primary goal of testing whether TFAM protects mtDNA from UVC damage and the impact this has on the mtDNA. While their data points to an intriguing model that TFAM acts as a sensor of damaged mtDNA, the validation of this model requires further investigation to make the model more convincing. This is likely warranted for a follow-up study. Also, the biological impact of this compaction, such as altering transcription levels, is not clear in this study.*

We have updated wording in the Abstract, Introduction, and elsewhere in the text (as detailed in other portions of our response) to make as explicit and clear as possible which results are supported by the *in vitro* versus *in vivo* data, and which parts are conclusions supported by the data versus hypothesized models to be tested in future work.

#### *Impact and utility of the methods:*

*This work advances our understanding of how mitochondria manage UVC genome damage and proposes a structural mechanism for damage "sensing" independent of canonical repair. The methodology, including the custom TFAM DNA chip, will be broadly useful to the scientific community.*

#### *Context:*

*The study supports a model in which mitochondrial genome integrity is maintained not only by repair factors, but also by selective sequestration or removal of damaged*

genomes. The demonstration that TFAM compaction correlates with damage rather than protection reframes an interesting role in mtDNA quality control.

**Reviewer #2 (Public review):**

**Summary:**

King et al. present several sets of experiments aimed to address the potential impact of UV irradiation on human mitochondrial DNA as well as the possible role of mitochondrial TFAM protein in handling UV-irradiated mitochondrial genomes. The carefully worded conclusion derived from the results of experiments performed with human HeLa cells, in vitro small plasmid DNA, with PCR-generated human mitochondrial DNA, and with UV-irradiated small oligonucleotides is presented in the title of the manuscript: "UV irradiation alters TFAM binding to mitochondrial DNA". The authors also interpret results of somewhat unconnected experimental approaches to speculate that "TFAM is a potential DNA damage sensing protein in that it promotes UVC-dependent conformational changes in the [mitochondrial] nucleoids, making them more compact." They further propose that such a proposed compaction triggers the removal of UV-damaged mitochondrial genomes as well as facilitates replication of undamaged mitochondrial genomes.

**Strengths:**

(1) The authors presented convincing evidence that a very high dose (1500 J/m<sup>2</sup>) of UVC applied to oligonucleotides covering the entire mitochondrial DNA genome alleviates sequence specificity of TFAM binding (Figure 3). This high dose was sufficient to cause UV lesions in a large fraction of individual oligonucleotides. The method was developed in the lab of one of the corresponding authors (reference 74) and is technically well-refined. This result can be published as is or in combination with other data.

(2) The manuscript also presents AFM evidence (Figure 4) that TFAM, which was long known to facilitate compaction of the mitochondrial genome (Alam et al., 2003; PMID 12626705 and follow-up citations), causes in vitro compaction of a small pUC19 plasmid and that approximately 3 UVC lesions per plasmid molecule result in a slight, albeit detectable, increase in TFAM compaction of the plasmid. Both results can be discussed in line with a possible extrapolation to in vivo phenomena, but such a discussion should include a clear statement that no in vivo support was provided within the set of experiments presented in the manuscript.

We thank this reviewer for their careful reading and interpretation of the manuscript. We agree that discussion of in vivo implications and extrapolations need clear statements indicating where there is not currently in vivo support. We have updated the text throughout the paper to include this.

**Weaknesses:**

Besides the experiments presented in Figures 3 and 4, other results do not either support or contradict the speculation that TFAM can play a protective role, eliminating mitochondrial genomes with bulky lesions by way of excessive compaction and removing damaged genomes from the in vivo pool.

**To specify these weaknesses:**

(1) Figure 1 - presents evidence that UVC causes a reduction in the number of mitochondrial spots in cells. The role of TFAM is not assessed.

We are working to understand the role of TFAM in vivo following UV irradiation, but believe that work should be included in follow up studies rather than this publication.

*(2) Figure 2 - presents evidence that UVC causes lesions in mitochondrial genomes in vivo, detectable by qPCR. No direct assessment of TFAM roles in damage repair or mitochondrial DNA turnover is assessed despite the statements in the title of Figure 2 or in associated text. Approximately 2-fold change in gene expression of TFAM and of the three other genes does not provide any reasonable support to suggestion about increased mitochondrial DNA turnover over multiple explanations on related to mitochondrial DNA maintenance.*

We agree and have updated the title of Figure 2 to better reflect the findings outlined in the figure as well as the text.

The new title is, “UVC causes mtDNA damage that decreases over time and is associated with upregulation of mtDNA replication genes, in the absence of apparent mitochondrial dysfunction.”

We agree that there are numerous mechanistic hypotheses that could explain the decrease in mtDNA damage over time. In Figure 1, we show that there is an overall decrease in mtDNA spots, and an increase in mtDNA-lysosome colocalization, suggestive of mtDNA degradation, which could serve to remove damaged genomes. One possibility is that TFAM is playing a role in the damage removal (but not repair per cell as these lesions are not repaired). Another is changes in mtDNA turnover via increasing the replication machinery in order the synthesize non-damaged mtDNA molecules to dilute out damage. These and other possibilities are not mutually exclusive. We have added text (pages 8-9) to make explicit that additional work will be required to distinguish these possibilities. We note that we have also added an additional experiment showing that TFAM knockdown affects mtDNA damage at baseline, as well as after UVC exposure (Figure 5J).

*(3) Figure 5. Shows that TFAM does not protect either mitochondrial nucleoids formed in vitro or mitochondrial DNA in vivo from UVC lesions as well as has no effect on in vivo repair of UV lesions.*

We agree that Figure 5 shows that TFAM does not protect DNA from UVC-induced lesions, and that a roughly 2-fold increase in TFAM protein does not alter damage reduction over time. We have added new data showing that *in vivo*, knockdown of TFAM results in an increase in baseline (control conditions) mtDNA damage, and also alters the rate of decrease of mtDNA damage over time after UVC (Figure 5J).

*(4) Figure 6: Based on the above analysis, the model of the role of TFAM in sensing mtDNA damage and elimination of damaged genomes in vivo appears unsupported.*

We have updated the legend for Figure 6 in which we outline our hypothesized role of TFAM in sensing mtDNA damage to ensure that readers know this has yet to be fully tested *in vivo*. We have also updated the Figure legend title from “proposed model” to “hypothesized model,” and changed the wording in the conclusion section (page 11) to highlight more clearly that this is a working model.

*(5) Additional concern about Figure 3 and relevant discussion: It is not clear if more uniform TFAM binding to UV irradiated oligonucleotides with varying sequence as compared to non-irradiated oligonucleotides can be explained by just overall reduced binding eliminating sequence specific peaks.*

We do not believe this is the case given the similar  $K_D$  values for the sequences tested. In our hands and in other publications (reviewed in PMID: 34440420), it has been well established that TFAM binds damaged DNA very well—essentially just as well as nondamaged DNA or better.



Additionally, a reduction in overall binding on these DNA arrays tends to make sequence specific peaks more apparent. We ran our experiments at both 30 nM and 300 nM TFAM specifically to be able to assess this question. The 300 nM data can be found in supplemental Figure S7. In this figure, we notice that the peaks appear more uniform at the high concentration (comparing Figure 3A to Figure S7A). That is presumably because there is so much more binding happening across the array that the peaks associated with the strongest binders become less pronounced. For the sake of brevity, we have not added this reasoning to the text, but are willing to do so if the Reviewers and Editor feel that it is important to include.

**Reviewer #3 (Public review):**

*Summary:*

*The study is grounded in the observations that mitochondrial DNA (mtDNA) exhibits a degree of resistance to mutagenesis under genotoxic stress. The manuscript focuses on the effects of UVC-induced DNA damage on TFAM-DNA binding in vitro and in cells. The authors demonstrate increased TFAM-DNA compaction following UVC irradiation in vitro based on high-throughput protein-DNA binding and atomic force microscopy (AFM) experiments. They did not observe a similar trend in fluorescence polarization assays. In cells, the authors found that UVC exposure upregulated TFAM, POLG, and POLRMT mRNA levels without affecting the mitochondrial membrane potential. Overexpressing TFAM in cells or varying TFAM concentration in reconstituted nucleoids did not alter the accumulation or disappearance of mtDNA damage. Based on their data, the authors proposed a plausible model that, following UVC-induced DNA damage, TFAM facilitates nucleoid compaction, which may serve to signal damage in the mitochondrial genome.*

*Strengths:*

*The presented data are solid, technically rigorous, and consistent with established literature findings. The experiments are well-executed, providing reliable evidence on the change of TFAM-DNA interactions following UVC irradiation. The proposed model may inspire future follow-up studies to further study the role of TFAM in sensing UVC-induced damage.*

*Weaknesses:*

*The manuscript could be further improved by refining specific interpretations and ensuring terminology aligns precisely with the data presented.*

*(1) In line 322, the claim of increased "nucleoid compaction" in cells should be removed, as there is a lack of direct cellular evidence. Given that non-DNA-bound TFAM is subject to protease digestion, it is uncertain to what extent the overexpressed TFAM actually integrates into and compacts mitochondrial nucleoids in the absence of supporting immunofluorescence data.*

We would like to thank this reviewer for their comments and suggestions. We feel these specific language changes have strengthened the interpretability of the text. The TFAM overexpression cells used in this experiment were given to us by Isaac et al., who demonstrated that when TFAM was overexpressed in this specific cell line, the nucleoids were indeed more compact, measured by Fiber-seq (Isaac et al., 2024; PMID: 38347148). We have removed the claim "increased compaction" from the section title, Figure 5 legend title, and from line 322 (now on page 8), and have also added an additional sentence to ensure the reader knows these cells have been shown to have presumed increased compaction by other groups.

(2) In lines 405 and 406, the authors should avoid equating TFAM overexpression with compaction in the cellular context unless the compaction is directly visualized or measured.

We have updated the text to ensure that it is clear that this was tested by other groups. We also changed the wording to “inaccessible (presumably compacted) nucleoids.” While we did not demonstrate altered compaction in our study, we think that based on the results from Isaac et al., it is likely that there was increased compaction. In addition, some readers might not have the context to make the connection between compaction and accessibility, so eliminating all reference to compaction could obscure the point.

(3) In lines 304 and 305 (and several other places throughout the manuscript), the authors use the term “removal rates”. A “removal rate” requires a direct comparison of accumulated lesion levels over a time course under different conditions. Given the complexity of UV-induced DNA damage—which involves both damage formation and potential removal via multiple pathways—a more accurate term that reflects the net result of these opposing processes is “accumulated DNA damage levels.” This terminology better reflects the final state measured and avoids implying a single, active ‘removal’ pathway without sufficient kinetic data.

We agree and have updated the language throughout the text as well as the results heading for this section.

(4) In line 357, the authors refer to the decrease in the total DNA damage level as “The removal of damaged mtDNA”. The decrease may be simply due to the turnover and resynthesis of non-damaged mtDNA molecules. The term “removal” may mislead the casual reader into interpreting the effect as an active repair/removal process.

We agree and have restructured this sentence for clarity. We do believe there is some removal happening, given the increase in mtDNA colocalization in lysosomes alongside decrease of mtDNA spots in our live cell imaging. We have written it to reflect the inclusion of removal and resynthesis of nondamaged mtDNA molecules (see pages 8-9).

#### **Recommendations for the authors:**

##### **Reviewing Editor Comments:**

*The reviewers appreciate the quality of the presented data but concur that they do not support the primary claims in the title and abstract. The reviewers also realize that in vivo evidence for the model would require extensive new experimentation that goes beyond a reasonable revision. The recommendation is to change the title and significantly revise text, figure titles and legends for transparency, and conclusions within results and discussion sections.*

We thank the editor and all the reviewers for their feedback. We have added additional experiments, updated text throughout the entire paper to ensure our claims are supported, and revised our title. We feel that the changes we have made have indeed made the paper stronger, more transparent, and that the evidence put forth in this paper provides support for all claims made.

##### **Reviewer #1 (Recommendations for the authors):**

(1) Clarify mitochondrial response kinetics by adding an intermediate (e.g., 12 hrs) recovery timepoint for transcriptional analysis to resolve when TFAM and replication genes are induced.

We have added additional timepoints of 12 and 18 hours following exposure in Figure 2. These results strengthen our finding that the nuclear transcriptional program supporting mtDNA replication appears to be activated prior to the nuclear transcriptional program supporting mitochondrial transcription, in that POLG and TFAM come up before POLRMT and ND1.

*(2) Strengthen functional readouts by assessing additional parameters of mitochondrial function to substantiate the claim that UVC does not impair mitochondrial performance.*

We have referenced our previously-published data on mtROS and added a measurement of ATP following UVC exposure in Figure 2.

*(3) Consider exploring whether mtDNA degradation occurs via mitophagy, nucleoid-phagy, or another pathway-potentially by using inhibitors or markers of these processes.*

While we agree that this is an important follow up question and are currently working on experiments to address this, those experiments are outside the scope of this manuscript.

*(4) Provide additional details for the high occupancy TFAM sites. Provide brief annotation or discussion of genomic regions showing strong TFAM binding under non-irradiated conditions that are lost during UVC treatment. This would be helpful to the field as a whole.*

We have updated our discussion section to include this.

*(5) Include or discuss a control using UVC irradiated pUC19 without TFAM to confirm that observed compaction categories are TFAM dependent rather than an UVC induced DNA distortion.*

We have added in a supplemental figure (Figure S16) containing comparison of area analysis of control pUC19 and UV-irradiated pUC19 and we have added associated text in the results section of the paper.

*(6) It would be interesting to explore the link between compaction to transcriptional output. In the TFAM overexpression model, the authors could measure expression of mtDNA encoded transcripts (e.g., ND1, COX1) to connect increased compaction with altered mitochondrial transcription.*

While we agree that understanding how the compactional status alters mitochondrial transcription is worthwhile, we believe this is beyond the scope of this paper. Furthermore, this connection has previously been shown by Bruser et al., 2021 (PMID: 34818548) who showed that more compact nucleoids are not undergoing active transcription. It will be interesting to see in future work if mtDNA damage drives changes in both compaction as well as transcriptional activity.

*(7) Clarify quantitative presentation in figure 2F to explicitly note whether the observed increase in fluorescence intensity was statistically insignificant and confirm that the assay sensitivity is sufficient to detect small potential changes. As presented it is not clear if there is a change.*

We have changed the presentation of Figure 2F. There is a slight increase in membrane potential at the 24-hour time point and we have made that clear in the text as well. We included FCCP as a (standard) positive control, for which we can detect the associated decrease in membrane potential for. While it is always possible that a very small decrease occurred that we were unable to detect, we note that none of the six UVC-exposed groups that we tested even trended towards a decrease in MMP, making it less likely that there was an effect that we simply lacked the power or sensitivity to detect.

(8) *It would be interesting if the authors can comment on whether TFAM induced compaction after UVC might shield mtDNA from other, repairable lesions (e.g., oxidative or alkylation damage), offering a broader context for this mechanism beyond just UVC.*

In theory, we believe this is possible. It will also be interesting to see if the increased compaction following UVC also protects or shields the mtDNA from other enzymatic processes, such as repair proteins that may be searching for repairable lesions such as oxidative or alkylation damage. In this case, it seems as though the increased compaction would prevent the repair from happening at genomes harboring damage.

In this study we show with our *in vitro* nucleoids that the increased compaction does not protect against UVC, but this is likely because UVC does not need physical access to the DNA in order to damage it, as the wavelengths of UVC (centered in this case at 254nm) are readily absorbed by proteins and thus can go right through the proteins. Currently, we know that increased compaction by TFAM makes the DNA inaccessible to the enzymes required to methylate DNA used in Fiber-seq (PMID: 38347148), but we do not know if the compaction is tight enough to prevent ROS or alkylating agents from damaging the DNA. We have updated text in the discussion on page 10 to highlight some of these ideas.

**Reviewer #2 (Recommendations for the authors):**

*Please, go over all display items and text and clarify details that can help readers to understand important specifics of the experiments. Examples are provided below:*

(1) *Abstract and Introduction - indicate species and cell line*

We have updated the text to include this information.

(2) *Table 1 "TFAM KD measurements"- title and footnotes are entirely cryptic. Please, clarify the experimental design, question(s) addressed and conclusions drawn from data.*

We have updated the title of Table 1 to "Binding of TFAM to array sequences, measured using fluorescence anisotropy," and clarified the footnotes to make sure it is clear which sequences were selected for AFM oligomerization experiments.

(3) *Figure 3 and Material and Methods - specify UVC dose.*

We have added this information to both the figure legend and the methods section.

(4) *Figure 4 - specify UVC dose.*

We have added this information to the figure legend.

(5) *Figure 5. Panel B indicate which band is TFAM and which is HA-tag; Indicate clearly which panel is showing in vivo or in vitro results.*

We have updated the figure to label the untagged TFAM and HA-tagged TFAM and changed the panel titles to specify if they are *in vivo* results.

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